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Platoon-based protocol interworking

Abstract

Disclosed are systems, apparatuses, processes, and computer-readable media for wireless communications. For example, a process may include receiving, at a network device (e.g., a vehicle or other type of network device), a plurality of messages using a first type of communication protocol, wherein the network device is part of a platoon including a plurality of network devices (e.g., a plurality of vehicles or other network device). The process may include generating, at the network device, a summary message summarizing the plurality of messages. The process may include transmitting the summary message using a second type of communication protocol.

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Background/Summary

FIELD
(1) The present disclosure generally relates to vehicle communications. For example, aspects of the present disclosure relate to a platoon-based protocol interworking solution (e.g., for cellular vehicle-to-everything (C-V2X) and dedicated short-range communications (DSRC)).
BACKGROUND
(2) Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting

communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

(3) These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long-Term Evolution (LTE) standard. Aspects of wireless communication may comprise direct communication between devices, such as in V2X, vehicle-to-vehicle (V2V), and/or device-to-device (D2D) communication. There exists a need for further improvements in V2X, V2V, and/or D2D technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

SUMMARY

(4) The following presents a simplified summary relating to one or more aspects disclosed herein. Thus, the following summary should not be considered an extensive overview relating to all contemplated aspects, nor should the following summary be considered to identify key or critical elements relating to all contemplated aspects or to delineate the scope associated with any particular aspect. Accordingly, the following summary has the sole purpose to present certain concepts relating to one or more aspects relating to the mechanisms disclosed herein in a simplified form to precede the detailed description presented below.

(5) Disclosed are systems, apparatuses, methods and computer-readable media for a platoon-based protocol interworking solution (e.g., for cellular vehicle-to-everything (C-V2X) and dedicated short-range communications (DSRC)). According to at least one example, a method is provided for wireless communications at a network device. The method includes: receiving, at the network device, a plurality of messages using a first type of communication protocol, wherein the network device is part of a platoon including a plurality of network devices; generating, at the network device, a summary message summarizing the plurality of messages; and transmitting the summary message using a second type of communication protocol.

(6) In another example, an apparatus for wireless communications is provided that includes at least one memory and at least one processor coupled to the at least one memory. The at least one processor is configured to: receive a plurality of messages using a first type of communication protocol, wherein the apparatus is part of a platoon including a plurality of network devices; generate a summary message summarizing the plurality of messages; and output the summary message for transmission using a second type of communication protocol.

(7) In another example, a non-transitory computer-readable medium of a computing device is provided that has stored thereon instructions that, when executed by one or more processors, cause the one or more processors to: receive a plurality of messages using a first type of communication protocol, wherein the computing device is part of a platoon including a plurality of network devices; generate a summary message summarizing the plurality of messages; and output the summary message for transmission using a second type of communication protocol.

(8) In another example, an apparatus for wireless communications is provided. The apparatus includes: receive a plurality of messages using a first type of communication protocol, wherein the

apparatus is part of a platoon including a plurality of network devices; generate a summary message summarizing the plurality of messages; and output the summary message for transmission using a second type of communication protocol.
(9) In some aspects, the apparatus is, or is part of, a vehicle (e.g., an automobile, truck, etc., or a component or system of an automobile, truck, etc.), a mobile device (e.g., a mobile telephone or so-called “smart phone” or other mobile device), a wearable device, an extended reality device (e.g., a virtual reality (VR) device, an augmented reality (AR) device, or a mixed reality (MR) device), a personal computer, a laptop computer, a server computer, a robotics device, or other device. In some aspects, the apparatus includes radio detection and ranging (radar) for capturing radio frequency (RF) signals. In some aspects, the apparatus includes one or more light detection and ranging (LIDAR) sensors, radar sensors, or other light-based sensors for capturing light-based (e.g., optical frequency) signals. In some aspects, the apparatus includes a camera or multiple cameras for capturing one or more images. In some aspects, the apparatus further includes a display for displaying one or more images, notifications, and/or other displayable data. In some aspects, the apparatuses described above can include one or more sensors, which can be used for determining a location of the apparatuses, a state of the apparatuses (e.g., a temperature, a humidity level, and/or other state), and/or for other purposes.
(10) This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended for use in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this patent, any or all drawings, and each claim.
(11) Other objects and advantages associated with the aspects disclosed herein will be apparent to those skilled in the art based on the accompanying drawings and detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Illustrative aspects of the present application are described in detail below with reference to the following figures:
(2) FIG. **1** is a diagram illustrating an example wireless communications system, in accordance with some aspects of the present disclosure.
(3) FIG. **2** is a diagram illustrating an example of a disaggregated base station architecture, which may be employed by the disclosed system for a platoon-based C-V2X and DSRC interworking solution, in accordance with some aspects of the present disclosure.
(4) FIG. **3** is a diagram illustrating an example of various user equipment (UEs) communicating over direct communication interfaces (e.g., a cellular based PC5 sidelink interface, 802.11p defined DSRC interface, or other direct interface) and wide area network (Uu) interfaces, in accordance with some aspects of the present disclosure.
(5) FIG. **4** is a block diagram illustrating an example of a computing system of a vehicle, in accordance with some aspects of the present disclosure.
(6) FIG. **5** is a block diagram illustrating an example of a computing system of a user device, in accordance with some aspects of the present disclosure.
(7) FIG. **6** is a diagram illustrating an example of devices involved in wireless communications (e.g., sidelink communications), in accordance with some aspects of the present disclosure.
(8) FIGS. **7**A-**7**D are diagrams illustrating examples of sensor-sharing for cooperative and automated driving systems, in accordance with some aspects of the present disclosure.
(9) FIG. **8** is a diagram illustrating an example of sensor-sharing for cooperative and automated driving systems, in accordance with some aspects of the present disclosure.
(10) FIG. **9** is a diagram illustrating an example of a system for sensor sharing in wireless

communications (e.g., C-V2X communications), in accordance with some aspects of the present disclosure.
(11) FIG. **10** is a diagram illustrating an example of a vehicle-based message (shown as a sensor-sharing message), in accordance with some aspects of the present disclosure.
(12) FIG. **11** is a diagram illustrating an example of a system employing a road side unit (RSU) as a bridge for communications between vehicles equipped for different types of communications (e.g., C-V2X and DSRC communications), in accordance with some aspects of the present disclosure.
(13) FIG. **12** is a diagram of the disclosed system for a platoon-based C-V2X and DSRC interworking solution, where some vehicles within the system are collaboratively negotiating the formation of a platoon, in accordance with some aspects of the present disclosure.
(14) FIGS. **13**A and **13**B are diagrams of the disclosed system for a platoon-based C-V2X and DSRC interworking solution that together illustrate an example of a platoon head operating as a bridge for communications between vehicles equipped for different types of communications (e.g., C-V2X and DSRC communications), in accordance with some aspects of the present disclosure.
(15) FIGS. **14**A and **14**B are diagrams of the disclosed system for a platoon-based C-V2X and DSRC interworking solution that together illustrate another example of a platoon head operating as a bridge for communications between vehicles equipped for different types of communications (e.g., C-V2X and DSRC communications), in accordance with some aspects of the present disclosure.
(16) FIG. **15** is a flow diagram illustrating an example of a process for platoon formation, in accordance with some aspects of the present disclosure.
(17) FIG. **16** is a flow chart illustrating an example of a process for wireless communications, according to some aspects of the present disclosure.
(18) FIG. **17** illustrates an example computing system, according to aspects of the disclosure.
DETAILED DESCRIPTION
(19) Certain aspects of this disclosure are provided below for illustration purposes. Alternate aspects may be devised without departing from the scope of the disclosure. Additionally, well-known elements of the disclosure will not be described in detail or will be omitted so as not to obscure the relevant details of the disclosure. Some of the aspects described herein can be applied independently and some of them may be applied in combination as would be apparent to those of skill in the art. In the following description, for the purposes of explanation, specific details are set forth in order to provide a thorough understanding of aspects of the application. However, it will be apparent that various aspects may be practiced without these specific details. The figures and description are not intended to be restrictive.
(20) The ensuing description provides example aspects only, and is not intended to limit the scope, applicability, or configuration of the disclosure. Rather, the ensuing description of the example aspects will provide those skilled in the art with an enabling description for implementing an example aspect. It should be understood that various changes may be made in the function and arrangement of elements without departing from the spirit and scope of the application as set forth in the appended claims.
(21) The terms “exemplary” and/or “example” are used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” and/or “example” is not necessarily to be construed as preferred or advantageous over other aspects. Likewise, the term “aspects of the disclosure” does not require that all aspects of the disclosure include the discussed feature, advantage or mode of operation.
(22) Wireless communications systems are deployed to provide various telecommunication services, including telephony, video, data, messaging, broadcasts, among others. Wireless communications systems have developed through various generations. A 5G mobile standard calls for higher data transfer speeds, greater numbers of connections, and better coverage, among other improvements. The 5G standard (also referred to as “New Radio” or “NR”), according to the Next

Generation Mobile Networks Alliance, is designed to provide data rates of several tens of megabits per second to each of tens of thousands of users.

(23) Vehicles are an example of devices or systems that can include wireless communications capabilities. For example, vehicles (e.g., automotive vehicles, autonomous vehicles, aircraft, maritime vessels, among others) can communicate with other vehicles and/or with other devices that have wireless communications capabilities. Wireless vehicle communication systems encompass vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), and vehicle-to-pedestrian (V2P) communications, which are all collectively referred to as vehicle-to-everything (V2X) communications. V2X communications is a vehicular communication system that supports the wireless transfer of information from a vehicle to other entities (e.g., other vehicles, pedestrians with smart phones, and/or other traffic infrastructure) located within the traffic system that may affect the vehicle. The main purpose of the V2X technology is to improve road safety, fuel savings, and traffic efficiency.

(24) In a V2X communication system, information is transmitted from vehicle sensors (and other sources) through wireless links to allow the information to be communicated to other vehicles, pedestrians, and/or traffic infrastructure. The information may be transmitted using one or more vehicle-based messages, such as C-V2X messages, which can include Sensor Data Sharing Messages (SDSMs), Basic Safety Messages (BSMs), Cooperative Awareness Messages (CAMs), Collective Perception Messages (CPMs), and/or other type of message. By sharing this information with other vehicles, the V2X technology improves vehicle (and driver) awareness of potential dangers to help reduce collisions with other vehicles and entities. In addition, the V2X technology enhances traffic efficiency by providing traffic warnings to vehicles of potential upcoming road dangers and obstacles such that vehicles may choose alternative traffic routes.

(25) As previously mentioned, the V2X technology includes V2V communications, which can also be referred to as peer-to-peer communications. V2V communications allows for vehicles to directly wireless communicate with each other while on the road. With V2V communications, vehicles can gain situational awareness by receiving information regarding upcoming road dangers (e.g., unforeseen oncoming vehicles, accidents, and road conditions) from the other vehicles.

(26) The IEEE 802.11p Standard supports (uses) a dedicated short-range communications (DSRC) interface for V2X wireless communications. Characteristics of the IEEE 802.11p based DSRC interface include low latency and the use of the unlicensed 5.9 Gigahertz (GHz) frequency band. Cellular V2X (C-V2X) was adopted as an alternative to using the IEEE 802.11p based DSRC interface for the wireless communications. The 5G Automotive Association (5GAA) supports the use of C-V2X technology. In some cases, the C-V2X technology uses Long-Term Evolution (LTE) as the underlying technology, and the C-V2X functionalities are based on the LTE technology. C-V2X includes a plurality of operational modes. One of the operational modes allows for direct wireless communication between vehicles over the LTE sidelink PC5 interface. Similar to the IEEE 802.11p based DSRC interface, the LTE C-V2X sidelink PC5 interface operates over the 5.9 GHz frequency band. Vehicle-based messages, such as BSMs and CAMs, which are application layer messages, are designed to be wirelessly broadcasted over the 802.11p based DSRC interface and the LTE C-V2X sidelink PC5 interface.

(27) Currently, many commercially available vehicles in the United States are equipped for C-V2X communications to utilize the C-V2X sidelink PC5 interface for communications with other vehicles similarly equipped for C-V2X communications (e.g., to transmit and/or receive vehicle-based messages, such as safety and awareness messages and warnings). Some of these vehicles in the United States are also equipped for DSRC communications to utilize the 802.11p based DSRC interface for communications with other vehicles similarly equipped for DSRC communications (e.g., to transmit and/or receive vehicle-based messages). As such, these vehicles with a dual communications capability are said to have a heterogeneous C-V2X/DSRC communications capability.

(28) Conversely, in Europe for example, many commercially available vehicles are only equipped for DSRC communications to utilize the 802.11p based DSRC interface for communications with other vehicles similarly equipped for DSRC communications (e.g., to transmit and/or receive vehicle-based messages). Vehicles that are only equipped for DSRC communications are not able to directly communicate with vehicles that are not equipped for DSRC communications. As such, vehicle-based messages (e.g., safety, awareness, and/or warning messages) generated by vehicles that are only equipped for DSRC communications cannot easily be received by vehicles that are only equipped for C-V2X communications, and vice versa.
(29) In some aspects of the present disclosure, systems, apparatuses, methods (also referred to as processes), and computer-readable media (collectively referred to herein as "systems and techniques") are described herein for providing a platoon-based protocol interworking solution (e.g., for C-V2X and DSRC interworking). For example, the systems and techniques can provide a solution for platoon-based protocol interworking among computing devices configured to communicate using only a first type of communication protocol (e.g., C-V2X), at least one computing device configured to communicate using only a second type of communication protocol (e.g., DSRC), and at least one computing device configured to communicate using both the first type of communication protocol and the second type of communication protocol (e.g., C-V2X and DSRC). While aspects described herein reference a vehicle as an illustrative example of a computing devices, the systems and techniques described herein may apply to other types of computing devices.
(30) In one or more aspects of the present disclosure, a plurality of vehicles within a grouping (e.g., located proximate to one another within an area or zone of a designated size) that are equipped to communicate using at least the first type of communication protocol (e.g., C-V2X) may form a platoon. The plurality of vehicles may negotiate amongst themselves to identify one of the vehicles within (or included in) the platoon to be the platoon head. In one example, a restriction or requirement for a vehicle to be determined as the platoon head is that the vehicle has heterogeneous communication capability using the first and second types of communication protocols (e.g., the vehicle has both C-V2X and DSRC communications capability). In addition to the heterogenous communication capability, in some cases, a vehicle can be determined to be the platoon head based on a location of the vehicle relative to locations of other vehicles in the platoon, such as if the vehicle selected as the platoon head is centrally located within the plurality of vehicles making up the platoon such that the platoon head vehicle is located within communications range of all of the other vehicles within the platoon.
(31) Because the platoon head has heterogeneous communications capability (e.g., the capability to transmit/receive messages using both C-V2X and DSRC communication protocols), the platoon head can operate as a bridge for communications (e.g., communications of vehicle-based messages, such as safety and awareness messages and warnings) between vehicles within the platoon (which are equipped to use at least the first type of communication protocol, such as C-V2X communications) and other vehicles not within the platoon that are equipped only for the second type of communication protocol (e.g., DSRC communications).
(32) Additional aspects of the present disclosure are described in more detail below.
(33) As used herein, the terms "user equipment" (UE) and "network entity" are not intended to be specific or otherwise limited to any particular radio access technology (RAT), unless otherwise noted. In general, a UE may be any wireless communication device (e.g., a mobile phone, router, tablet computer, laptop computer, and/or tracking device, etc.), wearable (e.g., smartwatch, smart-glasses, wearable ring, and/or an extended reality (XR) device such as a virtual reality (VR) headset, an augmented reality (AR) headset or glasses, or a mixed reality (MR) headset), vehicle (e.g., automobile, motorcycle, bicycle, etc.), and/or Internet of Things (IoT) device, etc., used by a user to communicate over a wireless communications network. A UE may be mobile or may (e.g., at certain times) be stationary, and may communicate with a radio access network (RAN). As used

herein, the term "UE" may be referred to interchangeably as an "access terminal" or "AT," a "client device," a "wireless device," a "subscriber device," a "subscriber terminal," a "subscriber station," a "user terminal" or "UT," a "mobile device," a "mobile terminal," a "mobile station," or variations thereof. Generally, UEs can communicate with a core network via a RAN, and through the core network the UEs can be connected with external networks such as the Internet and with other UEs. Of course, other mechanisms of connecting to the core network and/or the Internet are also possible for the UEs, such as over wired access networks, wireless local area network (WLAN) networks (e.g., based on IEEE 802.11 communication standards, etc.) and so on.

(34) In some cases, a network entity can be implemented in an aggregated or monolithic base station or server architecture, or alternatively, in a disaggregated base station or server architecture, and may include one or more of a central unit (CU), a distributed unit (DU), a radio unit (RU), a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC. In some cases, a network entity can include a server device, such as a Multi-access Edge Compute (MEC) device. A base station or server (e.g., with an aggregated/monolithic base station architecture or disaggregated base station architecture) may operate according to one of several RATs in communication with UEs, road side units (RSUs), and/or other devices depending on the network in which it is deployed, and may be alternatively referred to as an access point (AP), a network node, a NodeB (NB), an evolved NodeB (eNB), a next generation eNB (ng-eNB), a New Radio (NR) Node B (also referred to as a gNB or gNodeB), etc. A base station may be used primarily to support wireless access by UEs, including supporting data, voice, and/or signaling connections for the supported UEs. In some systems, a base station may provide edge node signaling functions while in other systems it may provide additional control and/or network management functions. A communication link through which UEs can send signals to a base station is called an uplink (UL) channel (e.g., a reverse traffic channel, a reverse control channel, an access channel, etc.). A communication link through which the base station can send signals to UEs is called a downlink (DL) or forward link channel (e.g., a paging channel, a control channel, a broadcast channel, or a forward traffic channel, etc.). The term traffic channel (TCH), as used herein, can refer to either an uplink, reverse or downlink, and/or a forward traffic channel.

(35) The term "network entity" or "base station" (e.g., with an aggregated/monolithic base station architecture or disaggregated base station architecture) may refer to a single physical TRP or to multiple physical TRPs that may or may not be co-located. For example, where the term "network entity" or "base station" refers to a single physical TRP, the physical TRP may be an antenna of the base station corresponding to a cell (or several cell sectors) of the base station. Where the term "network entity" or "base station" refers to multiple co-located physical TRPs, the physical TRPs may be an array of antennas (e.g., as in a multiple-input multiple-output (MIMO) system or where the base station employs beamforming) of the base station. Where the term "base station" refers to multiple non-co-located physical TRPs, the physical TRPs may be a distributed antenna system (DAS) (a network of spatially separated antennas connected to a common source via a transport medium) or a remote radio head (RRH) (a remote base station connected to a serving base station). Alternatively, the non-co-located physical TRPs may be the serving base station receiving the measurement report from the UE and a neighbor base station whose reference radio frequency (RF) signals (or simply "reference signals") the UE is measuring. Because a TRP is the point from which a base station transmits and receives wireless signals, as used herein, references to transmission from or reception at a base station are to be understood as referring to a particular TRP of the base station.

(36) In some implementations that support positioning of UEs, a network entity or base station may not support wireless access by UEs (e.g., may not support data, voice, and/or signaling connections for UEs), but may instead transmit reference signals to UEs to be measured by the UEs, and/or may receive and measure signals transmitted by the UEs. Such a base station may be referred to as a positioning beacon (e.g., when transmitting signals to UEs) and/or as a location measurement unit

(e.g., when receiving and measuring signals from UEs).

(37) A roadside unit (RSU) is a device that can transmit and receive messages over a communications link or interface (e.g., a cellular-based sidelink or PC5 interface, an 802.11 or WiFi™ based Dedicated Short Range Communication (DSRC) interface, and/or other interface) to and from one or more UEs, other RSUs, and/or base stations. An example of messages that can be transmitted and received by an RSU includes vehicle-to-everything (V2X) messages, which are described in more detail below. RSUs can be located on various transportation infrastructure systems, including roads, bridges, parking lots, toll booths, and/or other infrastructure systems. In some examples, an RSU can facilitate communication between UEs (e.g., vehicles, pedestrian user devices, and/or other UEs) and the transportation infrastructure systems. In some implementations, a RSU can be in communication with a server, base station, and/or other system that can perform centralized management functions.

(38) An RSU can communicate with a communications system of a UE. For example, an intelligent transport system (ITS) of a UE (e.g., a vehicle and/or other UE) can be used to generate and sign messages for transmission to an RSU and to validate messages received from an RSU. An RSU can communicate (e.g., over a PC5 interface, DSRC interface, etc.) with vehicles traveling along a road, bridge, or other infrastructure system in order to obtain traffic-related data (e.g., time, speed, location, etc. of the vehicle). In some cases, in response to obtaining the traffic-related data, the RSU can determine or estimate traffic congestion information (e.g., a start of traffic congestion, an end of traffic congestion, etc.), a travel time, and/or other information for a particular location. In some examples, the RSU can communicate with other RSUs (e.g., over a PC5 interface, DSRC interface, etc.) in order to determine the traffic-related data. The RSU can transmit the information (e.g., traffic congestion information, travel time information, and/or other information) to other vehicles, pedestrian UEs, and/or other UEs. For example, the RSU can broadcast or otherwise transmit the information to any UE (e.g., vehicle, pedestrian UE, etc.) that is in a coverage range of the RSU.

(39) A radio frequency signal or “RF signal” comprises an electromagnetic wave of a given frequency that transports information through the space between a transmitter and a receiver. As used herein, a transmitter may transmit a single “RF signal” or multiple “RF signals” to a receiver. However, the receiver may receive multiple “RF signals” corresponding to each transmitted RF signal due to the propagation characteristics of RF signals through multipath channels. The same transmitted RF signal on different paths between the transmitter and receiver may be referred to as a “multipath” RF signal. As used herein, an RF signal may also be referred to as a “wireless signal” or simply a “signal” where it is clear from the context that the term “signal” refers to a wireless signal or an RF signal.

(40) According to various aspects, FIG. **1** illustrates an exemplary wireless communications system **100**. The wireless communications system **100** (which may also be referred to as a wireless wide area network (WWAN)) can include various base stations **102** and various UEs **104**. In some aspects, the base stations **102** may also be referred to as “network entities” or “network nodes.” One or more of the base stations **102** can be implemented in an aggregated or monolithic base station architecture. Additionally or alternatively, one or more of the base stations **102** can be implemented in a disaggregated base station architecture, and may include one or more of a central unit (CU), a distributed unit (DU), a radio unit (RU), a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC), or a Non-Real Time (Non-RT) RIC. The base stations **102** can include macro cell base stations (high power cellular base stations) and/or small cell base stations (low power cellular base stations). In an aspect, the macro cell base station may include eNBs and/or ng-eNBs where the wireless communications system **100** corresponds to a long-term evolution (LTE) network, or gNBs where the wireless communications system **100** corresponds to a NR network, or a combination of both, and the small cell base stations may include femtocells, picocells, microcells, etc.

(41) The base stations **102** may collectively form a RAN and interface with a core network **170** (e.g., an evolved packet core (EPC) or a 5G core (5GC)) through backhaul links **122**, and through the core network **170** to one or more location servers **172** (which may be part of core network **170** or may be external to core network **170**). In addition to other functions, the base stations **102** may perform functions that relate to one or more of transferring user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, RAN sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may communicate with each other directly or indirectly (e.g., through the EPC or 5GC) over backhaul links **134**, which may be wired and/or wireless.
(42) The base stations **102** may wirelessly communicate with the UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area **110**. In an aspect, one or more cells may be supported by a base station **102** in each coverage area **110**. A "cell" is a logical communication entity used for communication with a base station (e.g., over some frequency resource, referred to as a carrier frequency, component carrier, carrier, band, or the like), and may be associated with an identifier (e.g., a physical cell identifier (PCI), a virtual cell identifier (VCI), a cell global identifier (CGI)) for distinguishing cells operating via the same or a different carrier frequency. In some cases, different cells may be configured according to different protocol types (e.g., machine-type communication (MTC), narrowband IoT (NB-IoT), enhanced mobile broadband (eMBB), or others) that may provide access for different types of UEs. Because a cell is supported by a specific base station, the term "cell" may refer to either or both of the logical communication entity and the base station that supports it, depending on the context. In addition, because a TRP is typically the physical transmission point of a cell, the terms "cell" and "TRP" may be used interchangeably. In some cases, the term "cell" may also refer to a geographic coverage area of a base station (e.g., a sector), insofar as a carrier frequency can be detected and used for communication within some portion of geographic coverage areas **110**.
(43) While neighboring macro cell base station **102** geographic coverage areas **110** may partially overlap (e.g., in a handover region), some of the geographic coverage areas **110** may be substantially overlapped by a larger geographic coverage area **110**. For example, a small cell base station **102'** may have a coverage area **110'** that substantially overlaps with the coverage area **110** of one or more macro cell base stations **102**. A network that includes both small cell and macro cell base stations may be known as a heterogeneous network. A heterogeneous network may also include home eNBs (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG).
(44) The communication links **120** between the base stations **102** and the UEs **104** may include uplink (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use MIMO antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links **120** may be through one or more carrier frequencies. Allocation of carriers may be asymmetric with respect to downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink).
(45) The wireless communications system **100** may further include a WLAN AP **150** in communication with WLAN stations (STAs) **152** via communication links **154** in an unlicensed frequency spectrum (e.g., 5 Gigahertz (GHz)). When communicating in an unlicensed frequency spectrum, the WLAN STAs **152** and/or the WLAN AP **150** may perform a clear channel assessment (CCA) or listen before talk (LBT) procedure prior to communicating in order to determine whether the channel is available. In some examples, the wireless communications system **100** can include devices (e.g., UEs, etc.) that communicate with one or more UEs **104**, base stations **102**, APs **150**,

etc. utilizing the ultra-wideband (UWB) spectrum. The UWB spectrum can range from 3.1 to 10.5 GHz.

(46) The small cell base station **102′** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell base station **102′** may employ LTE or NR technology and use the same 5 GHz unlicensed frequency spectrum as used by the WLAN AP **150**. The small cell base station **102′**, employing LTE and/or 5G in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network. NR in unlicensed spectrum may be referred to as NR-U. LTE in an unlicensed spectrum may be referred to as LTE-U, licensed assisted access (LAA), or MulteFire.

(47) The wireless communications system **100** may further include a millimeter wave (mmW) base station **180** that may operate in mmW frequencies and/or near mmW frequencies in communication with a UE **182**. The mmW base station **180** may be implemented in an aggregated or monolithic base station architecture, or alternatively, in a disaggregated base station architecture (e.g., including one or more of a CU, a DU, a RU, a Near-RT RIC, or a Non-RT RIC). Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in this band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW and/or near mmW radio frequency band have high path loss and a relatively short range. The mmW base station **180** and the UE **182** may utilize beamforming (transmit and/or receive) over an mmW communication link **184** to compensate for the extremely high path loss and short range. Further, it will be appreciated that in alternative configurations, one or more base stations **102** may also transmit using mmW or near mmW and beamforming. Accordingly, it will be appreciated that the foregoing illustrations are merely examples and should not be construed to limit the various aspects disclosed herein.

(48) Transmit beamforming is a technique for focusing an RF signal in a specific direction. Traditionally, when a network node or entity (e.g., a base station) broadcasts an RF signal, it broadcasts the signal in all directions (omni-directionally). With transmit beamforming, the network node determines where a given target device (e.g., a UE) is located (relative to the transmitting network node) and projects a stronger downlink RF signal in that specific direction, thereby providing a faster (in terms of data rate) and stronger RF signal for the receiving device(s). To change the directionality of the RF signal when transmitting, a network node can control the phase and relative amplitude of the RF signal at each of the one or more transmitters that are broadcasting the RF signal. For example, a network node may use an array of antennas (referred to as a “phased array” or an “antenna array”) that creates a beam of RF waves that can be “steered” to point in different directions, without actually moving the antennas. Specifically, the RF current from the transmitter is fed to the individual antennas with the correct phase relationship so that the radio waves from the separate antennas add together to increase the radiation in a desired direction, while canceling to suppress radiation in undesired directions.

(49) Transmit beams may be quasi-collocated, meaning that they appear to the receiver (e.g., a UE) as having the same parameters, regardless of whether or not the transmitting antennas of the network node themselves are physically collocated. In NR, there are four types of quasi-collocation (QCL) relations. Specifically, a QCL relation of a given type means that certain parameters about a second reference RF signal on a second beam can be derived from information about a source reference RF signal on a source beam. Thus, if the source reference RF signal is QCL Type A, the receiver can use the source reference RF signal to estimate the Doppler shift, Doppler spread, average delay, and delay spread of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type B, the receiver can use the source reference RF signal to estimate the Doppler shift and Doppler spread of a second reference RF signal transmitted on the

same channel. If the source reference RF signal is QCL Type C, the receiver can use the source reference RF signal to estimate the Doppler shift and average delay of a second reference RF signal transmitted on the same channel. If the source reference RF signal is QCL Type D, the receiver can use the source reference RF signal to estimate the spatial receive parameter of a second reference RF signal transmitted on the same channel.

(50) In receiving beamforming, the receiver uses a receive beam to amplify RF signals detected on a given channel. For example, the receiver can increase the gain setting and/or adjust the phase setting of an array of antennas in a particular direction to amplify (e.g., to increase the gain level of) the RF signals received from that direction. Thus, when a receiver is said to beamform in a certain direction, it means the beam gain in that direction is high relative to the beam gain along other directions, or the beam gain in that direction is the highest compared to the beam gain of other beams available to the receiver. This results in a stronger received signal strength, (e.g., reference signal received power (RSRP), reference signal received quality (RSRQ), signal-to-interference-plus-noise ratio (SINR), etc.) of the RF signals received from that direction.

(51) Receive beams may be spatially related. A spatial relation means that parameters for a transmit beam for a second reference signal can be derived from information about a receive beam for a first reference signal. For example, a UE may use a particular receive beam to receive one or more reference downlink reference signals (e.g., positioning reference signals (PRS), tracking reference signals (TRS), phase tracking reference signal (PTRS), cell-specific reference signals (CRS), channel state information reference signals (CSI-RS), primary synchronization signals (PSS), secondary synchronization signals (SSS), synchronization signal blocks (SSBs), etc.) from a network node or entity (e.g., a base station). The UE can then form a transmit beam for sending one or more uplink reference signals (e.g., uplink positioning reference signals (UL-PRS), sounding reference signal (SRS), demodulation reference signals (DMRS), PTRS, etc.) to that network node or entity (e.g., a base station) based on the parameters of the receive beam.

(52) Note that a “downlink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a network node or entity (e.g., a base station) is forming the downlink beam to transmit a reference signal to a UE, the downlink beam is a transmit beam. If the UE is forming the downlink beam, however, it is a receive beam to receive the downlink reference signal. Similarly, an “uplink” beam may be either a transmit beam or a receive beam, depending on the entity forming it. For example, if a network node or entity (e.g., a base station) is forming the uplink beam, it is an uplink receive beam, and if a UE is forming the uplink beam, it is an uplink transmit beam.

(53) In 5G, the frequency spectrum in which wireless network nodes or entities (e.g., base stations **102**/**180**, UEs **104**/**182**) operate is divided into multiple frequency ranges, FR1 (from 450 to 6000 Megahertz (MHz)), FR2 (from 24250 to 52600 MHz), FR3 (above 52600 MHz), and FR4 (between FR1 and FR2). In a multi-carrier system, such as 5G, one of the carrier frequencies is referred to as the “primary carrier” or “anchor carrier” or “primary serving cell” or “PCell,” and the remaining carrier frequencies are referred to as “secondary carriers” or “secondary serving cells” or “SCells.” In carrier aggregation, the anchor carrier is the carrier operating on the primary frequency (e.g., FR1) utilized by a UE **104**/**182** and the cell in which the UE **104**/**182** either performs the initial radio resource control (RRC) connection establishment procedure or initiates the RRC connection re-establishment procedure. The primary carrier carries all common and UE-specific control channels, and may be a carrier in a licensed frequency (however, this is not always the case). A secondary carrier is a carrier operating on a second frequency (e.g., FR2) that may be configured once the RRC connection is established between the UE **104** and the anchor carrier and that may be used to provide additional radio resources. In some cases, the secondary carrier may be a carrier in an unlicensed frequency. The secondary carrier may contain only necessary signaling information and signals, for example, those that are UE-specific may not be present in the secondary carrier, since both primary uplink and downlink carriers are typically UE-specific. This means that

different UEs **104**/**182** in a cell may have different downlink primary carriers. The same is true for the uplink primary carriers. The network is able to change the primary carrier of any UE **104**/**182** at any time. This is done, for example, to balance the load on different carriers. Because a "serving cell" (whether a PCell or an SCell) corresponds to a carrier frequency and/or component carrier over which some base station is communicating, the term "cell," "serving cell," "component carrier," "carrier frequency," and the like can be used interchangeably.
(54) For example, still referring to FIG. **1**, one of the frequencies utilized by the macro cell base stations **102** may be an anchor carrier (or "PCell") and other frequencies utilized by the macro cell base stations **102** and/or the mmW base station **180** may be secondary carriers ("SCells"). In carrier aggregation, the base stations **102** and/or the UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100 MHz) bandwidth per carrier up to a total of Yx MHz (x component carriers) for transmission in each direction. The component carriers may or may not be adjacent to each other on the frequency spectrum. Allocation of carriers may be asymmetric with respect to the downlink and uplink (e.g., more or less carriers may be allocated for downlink than for uplink). The simultaneous transmission and/or reception of multiple carriers enables the UE **104**/**182** to significantly increase its data transmission and/or reception rates. For example, two 20 MHz aggregated carriers in a multi-carrier system would theoretically lead to a two-fold increase in data rate (i.e., 40 MHz), compared to that attained by a single 20 MHz carrier.
(55) In order to operate on multiple carrier frequencies, a base station **102** and/or a UE **104** is equipped with multiple receivers and/or transmitters. For example, a UE **104** may have two receivers, "Receiver 1" and "Receiver 2," where "Receiver 1" is a multi-band receiver that can be tuned to band (i.e., carrier frequency) 'X' or band 'Y,' and "Receiver 2" is a one-band receiver tunable to band 'Z' only. In this example, if the UE **104** is being served in band 'X,' band 'X' would be referred to as the PCell or the active carrier frequency, and "Receiver 1" would need to tune from band 'X' to band 'Y' (an SCell) in order to measure band 'Y' (and vice versa). In contrast, whether the UE **104** is being served in band 'X' or band 'Y,' because of the separate "Receiver 2," the UE **104** can measure band 'Z' without interrupting the service on band 'X' or band 'Y.'
(56) The wireless communications system **100** may further include a UE **164** that may communicate with a macro cell base station **102** over a communication link **120** and/or the mmW base station **180** over an mmW communication link **184**. For example, the macro cell base station **102** may support a PCell and one or more SCells for the UE **164** and the mmW base station **180** may support one or more SCells for the UE **164**.
(57) The wireless communications system **100** may further include one or more UEs, such as UE **190**, that connects indirectly to one or more communication networks via one or more device-to-device (D2D) peer-to-peer (P2P) links (referred to as "sidelinks"). In the example of FIG. **1**, UE **190** has a D2D P2P link **192** with one of the UEs **104** connected to one of the base stations **102** (e.g., through which UE **190** may indirectly obtain cellular connectivity) and a D2D P2P link **194** with WLAN STA **152** connected to the WLAN AP **150** (through which UE **190** may indirectly obtain WLAN-based Internet connectivity). In an example, the D2D P2P links **192** and **194** may be supported with any well-known D2D RAT, such as LTE Direct (LTE-D), Wi-Fi Direct (Wi-Fi-D), Bluetooth®, and so on.
(58) FIG. **2** is a diagram illustrating an example of a disaggregated base station architecture, which may be employed by the disclosed system for a platoon-based C-V2X and DSRC interworking solution, in accordance with some examples. Deployment of communication systems, such as 5G NR systems, may be arranged in multiple manners with various components or constituent parts. In a 5G NR system, or network, a network node, a network entity, a mobility element of a network, a radio access network (RAN) node, a core network node, a network element, or a network equipment, such as a base station (BS), or one or more units (or one or more components) performing base station functionality, may be implemented in an aggregated or disaggregated

architecture. For example, a BS (such as a Node B (NB), evolved NB (eNB), NR BS, 5G NB, AP, a transmit receive point (TRP), or a cell, etc.) may be implemented as an aggregated base station (also known as a standalone BS or a monolithic BS) or a disaggregated base station.
(59) An aggregated base station may be configured to utilize a radio protocol stack that is physically or logically integrated within a single RAN node. A disaggregated base station may be configured to utilize a protocol stack that is physically or logically distributed among two or more units (such as one or more central or centralized units (CUs), one or more distributed units (DUs), or one or more radio units (RUs)). In some aspects, a CU may be implemented within a RAN node, and one or more DUs may be co-located with the CU, or alternatively, may be geographically or virtually distributed throughout one or multiple other RAN nodes. The DUs may be implemented to communicate with one or more RUs. Each of the CU, DU and RU also can be implemented as virtual units, i.e., a virtual central unit (VCU), a virtual distributed unit (VDU), or a virtual radio unit (VRU).
(60) Base station-type operation or network design may consider aggregation characteristics of base station functionality. For example, disaggregated base stations may be utilized in an integrated access backhaul (IAB) network, an open radio access network (O-RAN (such as the network configuration sponsored by the O-RAN Alliance)), or a virtualized radio access network (vRAN, also known as a cloud radio access network (C-RAN)). Disaggregation may include distributing functionality across two or more units at various physical locations, as well as distributing functionality for at least one unit virtually, which can enable flexibility in network design. The various units of the disaggregated base station, or disaggregated RAN architecture, can be configured for wired or wireless communication with at least one other unit.
(61) As previously mentioned, FIG. **2** shows a diagram illustrating an example disaggregated base station **201** architecture. The disaggregated base station **201** architecture may include one or more central units (CUs) **211** that can communicate directly with a core network **223** via a backhaul link, or indirectly with the core network **223** through one or more disaggregated base station units (such as a Near-Real Time (Near-RT) RAN Intelligent Controller (RIC) **227** via an E2 link, or a Non-Real Time (Non-RT) RIC **217** associated with a Service Management and Orchestration (SMO) Framework **207**, or both). A CU **211** may communicate with one or more distributed units (DUs) **231** via respective midhaul links, such as an F1 interface. The DUs **231** may communicate with one or more radio units (RUs) **241** via respective fronthaul links. The RUs **241** may communicate with respective UEs **221** via one or more RF access links. In some implementations, the UE **221** may be simultaneously served by multiple RUs **241**.
(62) Each of the units, i.e., the CUs **211**, the DUs **231**, the RUs **241**, as well as the Near-RT RICs **227**, the Non-RT RICs **217** and the SMO Framework **207**, may include one or more interfaces or be coupled to one or more interfaces configured to receive or transmit signals, data, or information (collectively, signals) via a wired or wireless transmission medium. Each of the units, or an associated processor or controller providing instructions to the communication interfaces of the units, can be configured to communicate with one or more of the other units via the transmission medium. For example, the units can include a wired interface configured to receive or transmit signals over a wired transmission medium to one or more of the other units. Additionally, the units can include a wireless interface, which may include a receiver, a transmitter or transceiver (such as an RF transceiver), configured to receive or transmit signals, or both, over a wireless transmission medium to one or more of the other units.
(63) In some aspects, the CU **211** may host one or more higher layer control functions. Such control functions can include radio resource control (RRC), packet data convergence protocol (PDCP), service data adaptation protocol (SDAP), or the like. Each control function can be implemented with an interface configured to communicate signals with other control functions hosted by the CU **211**. The CU **211** may be configured to handle user plane functionality (i.e., Central Unit-User Plane (CU-UP)), control plane functionality (i.e., Central Unit-Control Plane

(CU-CP)), or a combination thereof. In some implementations, the CU **211** can be logically split into one or more CU-UP units and one or more CU-CP units. The CU-UP unit can communicate bidirectionally with the CU-CP unit via an interface, such as the E1 interface when implemented in an O-RAN configuration. The CU **211** can be implemented to communicate with the DU **131**, as necessary, for network control and signaling.
(64) The DU **231** may correspond to a logical unit that includes one or more base station functions to control the operation of one or more RUs **241**. In some aspects, the DU **231** may host one or more of a radio link control (RLC) layer, a medium access control (MAC) layer, and one or more high physical (PHY) layers (such as modules for forward error correction (FEC) encoding and decoding, scrambling, modulation and demodulation, or the like) depending, at least in part, on a functional split, such as those defined by the 3.sup.rd Generation Partnership Project (3GPP). In some aspects, the DU **231** may further host one or more low PHY layers. Each layer (or module) can be implemented with an interface configured to communicate signals with other layers (and modules) hosted by the DU **231**, or with the control functions hosted by the CU **211**.
(65) Lower-layer functionality can be implemented by one or more RUs **241**. In some deployments, an RU **241**, controlled by a DU **231**, may correspond to a logical node that hosts RF processing functions, or low-PHY layer functions (such as performing fast Fourier transform (FFT), inverse FFT (iFFT), digital beamforming, physical random access channel (PRACH) extraction and filtering, or the like), or both, based at least in part on the functional split, such as a lower layer functional split. In such an architecture, the RU(s) **241** can be implemented to handle over the air (OTA) communication with one or more UEs **221**. In some implementations, real-time and non-real-time aspects of control and user plane communication with the RU(s) **241** can be controlled by the corresponding DU **231**. In some scenarios, this configuration can enable the DU(s) **231** and the CU **211** to be implemented in a cloud-based RAN architecture, such as a vRAN architecture.
(66) The SMO Framework **207** may be configured to support RAN deployment and provisioning of non-virtualized and virtualized network elements. For non-virtualized network elements, the SMO Framework **207** may be configured to support the deployment of dedicated physical resources for RAN coverage requirements which may be managed via an operations and maintenance interface (such as an O1 interface). For virtualized network elements, the SMO Framework **207** may be configured to interact with a cloud computing platform (such as an open cloud (O-Cloud) **291**) to perform network element life cycle management (such as to instantiate virtualized network elements) via a cloud computing platform interface (such as an O2 interface). Such virtualized network elements can include, but are not limited to, CUs **211**, DUs **231**, RUs **241** and Near-RT RICs **227**. In some implementations, the SMO Framework **207** can communicate with a hardware aspect of a 4G RAN, such as an open eNB (O-eNB) **213**, via an O1 interface. Additionally, in some implementations, the SMO Framework **207** can communicate directly with one or more RUs **241** via an O1 interface. The SMO Framework **207** also may include a Non-RT RIC **217** configured to support functionality of the SMO Framework **207**.
(67) The Non-RT RIC **217** may be configured to include a logical function that enables non-real-time control and optimization of RAN elements and resources, Artificial Intelligence/Machine Learning (AI/ML) workflows including model training and updates, or policy-based guidance of applications/features in the Near-RT RIC **227**. The Non-RT RIC **217** may be coupled to or communicate with (such as via an A1 interface) the Near-RT RIC **227**. The Near-RT RIC **227** may be configured to include a logical function that enables near-real-time control and optimization of RAN elements and resources via data collection and actions over an interface (such as via an E2 interface) connecting one or more CUs **211**, one or more DUs **231**, or both, as well as an O-eNB **213**, with the Near-RT RIC **227**.
(68) In some implementations, to generate AI/ML models to be deployed in the Near-RT RIC **227**, the Non-RT RIC **217** may receive parameters or external enrichment information from external servers. Such information may be utilized by the Near-RT RIC **227** and may be received at the

SMO Framework **207** or the Non-RT RIC **217** from non-network data sources or from network functions. In some examples, the Non-RT RIC **217** or the Near-RT RIC **227** may be configured to tune RAN behavior or performance. For example, the Non-RT RIC **217** may monitor long-term trends and patterns for performance and employ AI/ML models to perform corrective actions through the SMO Framework **207** (such as reconfiguration via O1) or via creation of RAN management policies (such as A1 policies).

(69) FIG. **3** illustrates examples of different communication mechanisms used by various UEs. In one example of sidelink communications, FIG. **3** illustrates a vehicle **304**, a vehicle **305**, and an RSU **303** communicating with each other using PC5, DSRC, or other device to device direct signaling interfaces. In addition, the vehicle **304** and the vehicle **305** may communicate with a base station **302** (shown as BS **302**) using a network (Uu) interface. The base station **302** can include a gNB in some examples. FIG. **3** also illustrates a user device **307** communicating with the base station **302** using a network (Uu) interface. As described below, functionalities can be transferred from a vehicle (e.g., vehicle **304**) to a user device (e.g., user device **307**) based on one or more characteristics or factors (e.g., temperature, humidity, etc.). In one illustrative example, V2X functionality can be transitioned from the vehicle **304** to the user device **307**, after which the user device **307** can communicate with other vehicles (e.g., vehicle **305**) over a PC5 interface (or other device to device direct interface, such as a DSRC interface), as shown in FIG. **3**.

(70) While FIG. **3** illustrates a particular number of vehicles (e.g., two vehicles **304** and **305**) communicating with each other and/or with RSU **303**, BS **302**, and/or user device **307**, the present disclosure is not limited thereto. For instance, tens or hundreds of such vehicles may be communicating with one another and/or with RSU **303**, BS **302**, and/or user device **307**. At any given point in time, each such vehicle, RSU **303**, BS **302**, and/or user device **307** may transmit various types of information as messages to other nearby vehicles resulting in each vehicle (e.g., vehicles **304** and/or **305**), RSU **303**, BS **302**, and/or user device **307** receiving hundreds or thousands of messages from other nearby vehicles, RSUs, base stations, and/or other UEs per second.

(71) While PC5 interfaces are shown in FIG. **3**, the various UEs (e.g., vehicles, user devices, etc.) and RSU(s) can communicate directly using any suitable type of direct interface, such as an 802.11 DSRC interface, a Bluetooth™ interface, and/or other interface. For example, a vehicle can communicate with a user device over a direct communications interface (e.g., using PC5 and/or DSRC), a vehicle can communicate with another vehicle over the direct communications interface, a user device can communicate with another user device over the direct communications interface, a UE (e.g., a vehicle, user device, etc.) can communicate with an RSU over the direct communications interface, an RSU can communicate with another RSU over the direct communications interface, and the like.

(72) FIG. **4** is a block diagram illustrating an example a vehicle computing system **450** of a vehicle **404**. The vehicle **404** is an example of a UE that can communicate with a network (e.g., an eNB, a gNB, a positioning beacon, a location measurement unit, and/or other network entity) over a Uu interface and with other UEs using V2X communications over a PC5 interface (or other device to device direct interface, such as a DSRC interface). As shown, the vehicle computing system **450** can include at least a power management system **451**, a control system **452**, an infotainment system **454**, an intelligent transport system (ITS) **455**, one or more sensor systems **456**, and a communications system **458**. In some cases, the vehicle computing system **450** can include or can be implemented using any type of processing device or system, such as one or more central processing units (CPUs), digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), application processors (APs), graphics processing units (GPUs), vision processing units (VPUs), Neural Network Signal Processors (NSPs), microcontrollers, dedicated hardware, any combination thereof, and/or other processing device or system.

(73) The control system **452** can be configured to control one or more operations of the vehicle **404**, the power management system **451**, the computing system **450**, the infotainment system **454**, the ITS **455**, and/or one or more other systems of the vehicle **404** (e.g., a braking system, a steering system, a safety system other than the ITS **455**, a cabin system, and/or other system). In some examples, the control system **452** can include one or more electronic control units (ECUs). An ECU can control one or more of the electrical systems or subsystems in a vehicle. Examples of specific ECUs that can be included as part of the control system **452** include an engine control module (ECM), a powertrain control module (PCM), a transmission control module (TCM), a brake control module (BCM), a central control module (CCM), a central timing module (CTM), among others. In some cases, the control system **452** can receive sensor signals from the one or more sensor systems **456** and can communicate with other systems of the vehicle computing system **450** to operate the vehicle **404**.
(74) The vehicle computing system **450** also includes a power management system **451**. In some implementations, the power management system **451** can include a power management integrated circuit (PMIC), a standby battery, and/or other components. In some cases, other systems of the vehicle computing system **450** can include one or more PMICs, batteries, and/or other components. The power management system **451** can perform power management functions for the vehicle **404**, such as managing a power supply for the computing system **450** and/or other parts of the vehicle. For example, the power management system **451** can provide a stable power supply in view of power fluctuations, such as based on starting an engine of the vehicle. In another example, the power management system **451** can perform thermal monitoring operations, such as by checking ambient and/or transistor junction temperatures. In another example, the power management system **451** can perform certain functions based on detecting a certain temperature level, such as causing a cooling system (e.g., one or more fans, an air conditioning system, etc.) to cool certain components of the vehicle computing system **450** (e.g., the control system **452**, such as one or more ECUs), shutting down certain functionalities of the vehicle computing system **450** (e.g., limiting the infotainment system **454**, such as by shutting off one or more displays, disconnecting from a wireless network, etc.), among other functions.
(75) The vehicle computing system **450** further includes a communications system **458**. The communications system **458** can include both software and hardware components for transmitting signals to and receiving signals from a network (e.g., a gNB or other network entity over a Uu interface) and/or from other UEs (e.g., to another vehicle or UE over a PC5 interface, WiFi interface (e.g., DSRC), Bluetooth™ interface, and/or other wireless and/or wired interface). For example, the communications system **458** is configured to transmit and receive information wirelessly over any suitable wireless network (e.g., a 3G network, 4G network, 5G network, WiFi network, Bluetooth™ network, and/or other network). The communications system **458** includes various components or devices used to perform the wireless communication functionalities, including an original equipment manufacturer (OEM) subscriber identity module (referred to as a SIM or SIM card) **460**, a user SIM **462**, and a modem **464**. While the vehicle computing system **450** is shown as having two SIMs and one modem, the computing system **450** can have any number of SIMs (e.g., one SIM or more than two SIMs) and any number of modems (e.g., one modem, two modems, or more than two modems) in some implementations.
(76) A SIM is a device (e.g., an integrated circuit) that can securely store an international mobile subscriber identity (IMSI) number and a related key (e.g., an encryption-decryption key) of a particular subscriber or user. The IMSI and key can be used to identify and authenticate the subscriber on a particular UE. The OEM SIM **460** can be used by the communications system **458** for establishing a wireless connection for vehicle-based operations, such as for conducting emergency-calling (eCall) functions, communicating with a communications system of the vehicle manufacturer (e.g., for software updates, etc.), among other operations. The OEM SIM **460** can be important for the OEM SIM to support critical services, such as eCall for making emergency calls

in the event of a car accident or other emergency. For instance, eCall can include a service that automatically dials an emergency number (e.g., "9-1-1" in the United States, "1-1-2" in Europe, etc.) in the event of a vehicle accident and communicates a location of the vehicle to the emergency services, such as a police department, fire department, etc.
(77) The user SIM **462** can be used by the communications system **458** for performing wireless network access functions in order to support a user data connection (e.g., for conducting phone calls, messaging, Infotainment related services, among others). In some cases, a user device of a user can connect with the vehicle computing system **450** over an interface (e.g., over PC5, Bluetooth™, WiFI™ (e.g., DSRC), a universal serial bus (USB) port, and/or other wireless or wired interface). Once connected, the user device can transfer wireless network access functionality from the user device to communications system **458** the vehicle, in which case the user device can cease performance of the wireless network access functionality (e.g., during the period in which the communications system **458** is performing the wireless access functionality). The communications system **458** can begin interacting with a base station to perform one or more wireless communication operations, such as facilitating a phone call, transmitting and/or receiving data (e.g., messaging, video, audio, etc.), among other operations. In such cases, other components of the vehicle computing system **450** can be used to output data received by the communications system **458**. For example, the infotainment system **454** (described below) can display video received by the communications system **458** on one or more displays and/or can output audio received by the communications system **458** using one or more speakers.
(78) A modem is a device that modulates one or more carrier wave signals to encode digital information for transmission, and demodulates signals to decode the transmitted information. The modem **464** (and/or one or more other modems of the communications system **458**) can be used for communication of data for the OEM SIM **460** and/or the user SIM **462**. In some examples, the modem **464** can include a 4G (or LTE) modem and another modem (not shown) of the communications system **458** can include a 5G (or NR) modem. In some examples, the communications system **458** can include one or more Bluetooth™ modems (e.g., for Bluetooth™ Low Energy (BLE) or other type of Bluetooth communications), one or more WiFi™ modems (e.g., for DSRC communications and/or other WiFi communications), wideband modems (e.g., an ultra-wideband (UWB) modem), any combination thereof, and/or other types of modems.
(79) In some cases, the modem **464** (and/or one or more other modems of the communications system **458**) can be used for performing V2X communications (e.g., with other vehicles for V2V communications, with other devices for D2D communications, with infrastructure systems for V2I communications, with pedestrian UEs for V2P communications, etc.). In some examples, the communications system **458** can include a V2X modem used for performing V2X communications (e.g., sidelink communications over a PC5 interface or DSRC interface), in which case the V2X modem can be separate from one or more modems used for wireless network access functions (e.g., for network communications over a network/Uu interface and/or sidelink communications other than V2X communications).
(80) In some examples, the communications system **458** can be or can include a telematics control unit (TCU). In some implementations, the TCU can include a network access device (NAD) (also referred to in some cases as a network control unit or NCU). The NAD can include the modem **464**, any other modem not shown in FIG. **4**, the OEM SIM **460**, the user SIM **462**, and/or other components used for wireless communications. In some examples, the communications system **458** can include a Global Navigation Satellite System (GNSS). In some cases, the GNSS can be part of the one or more sensor systems **456**, as described below. The GNSS can provide the ability for the vehicle computing system **450** to perform one or more location services, navigation services, and/or other services that can utilize GNSS functionality.
(81) In some cases, the communications system **458** can further include one or more wireless interfaces (e.g., including one or more transceivers and one or more baseband processors for each

wireless interface) for transmitting and receiving wireless communications, one or more wired interfaces (e.g., a serial interface such as a universal serial bus (USB) input, a lightening connector, and/or other wired interface) for performing communications over one or more hardwired connections, and/or other components that can allow the vehicle **404** to communicate with a network and/or other UEs.

(82) The vehicle computing system **450** can also include an infotainment system **454** that can control content and one or more output devices of the vehicle **404** that can be used to output the content. The infotainment system **454** can also be referred to as an in-vehicle infotainment (IVI) system or an In-car entertainment (ICE) system. The content can include navigation content, media content (e.g., video content, music or other audio content, and/or other media content), among other content. The one or more output devices can include one or more graphical user interfaces, one or more displays, one or more speakers, one or more extended reality devices (e.g., a VR, AR, and/or MR headset), one or more haptic feedback devices (e.g., one or more devices configured to vibrate a seat, steering wheel, and/or other part of the vehicle **404**), and/or other output device.

(83) In some examples, the computing system **450** can include the intelligent transport system (ITS) **455**. In some examples, the ITS **455** can be used for implementing V2X communications. For example, an ITS stack of the ITS **455** can generate V2X messages based on information from an application layer of the ITS. In some cases, the application layer can determine whether certain conditions have been met for generating messages for use by the ITS **455** and/or for generating messages that are to be sent to other vehicles (for V2V communications), to pedestrian UEs (for V2P communications), and/or to infrastructure systems (for V2I communications). In some cases, the communications system **458** and/or the ITS **455** can obtain car access network (CAN) information (e.g., from other components of the vehicle via a CAN bus). In some examples, the communications system **458** (e.g., a TCU NAD) can obtain the CAN information via the CAN bus and can send the CAN information to a PHY/MAC layer of the ITS **455**. The ITS **455** can provide the CAN information to the ITS stack of the ITS **455**. The CAN information can include vehicle related information, such as a heading of the vehicle, speed of the vehicle, breaking information, among other information. The CAN information can be continuously or periodically (e.g., every 1 millisecond (ms), every 10 ms, or the like) provided to the ITS **455**.

(84) The conditions used to determine whether to generate messages can be determined using the CAN information based on safety-related applications and/or other applications, including applications related to road safety, traffic efficiency, infotainment, business, and/or other applications. In one illustrative example, the ITS **455** can perform lane change assistance or negotiation. For instance, using the CAN information, the ITS **455** can determine that a driver of the vehicle **404** is attempting to change lanes from a current lane to an adjacent lane (e.g., based on a blinker being activated, based on the user veering or steering into an adjacent lane, etc.). Based on determining the vehicle **404** is attempting to change lanes, the ITS **455** can determine a lane-change condition has been met that is associated with a message to be sent to other vehicles that are nearby the vehicle in the adjacent lane. The ITS **455** can trigger the ITS stack to generate one or more messages for transmission to the other vehicles, which can be used to negotiate a lane change with the other vehicles. Other examples of applications include forward collision warning, automatic emergency breaking, lane departure warning, pedestrian avoidance or protection (e.g., when a pedestrian is detected near the vehicle **404**, such as based on V2P communications with a UE of the user), traffic sign recognition, among others.

(85) The ITS **455** can use any suitable protocol to generate messages (e.g., V2X messages). Examples of protocols that can be used by the ITS **455** include one or more Society of Automotive Engineering (SAE) standards, such as SAE J2735, SAE J2945, SAE J3161, and/or other standards, which are hereby incorporated by reference in their entirety and for all purposes.

(86) A security layer of the ITS **455** can be used to securely sign messages from the ITS stack that are sent to and verified by other UEs configured for V2X communications, such as other vehicles,

pedestrian UEs, and/or infrastructure systems. The security layer can also verify messages received from such other UEs. In some implementations, the signing and verification processes can be based on a security context of the vehicle. In some examples, the security context may include one or more encryption-decryption algorithms, a public and/or private key used to generate a signature using an encryption-decryption algorithm, and/or other information. For example, each ITS message generated by the ITS **455** can be signed by the security layer of the ITS **455**. The signature can be derived using a public key and an encryption-decryption algorithm. A vehicle, pedestrian UE, and/or infrastructure system receiving a signed message can verify the signature to make sure the message is from an authorized vehicle. In some examples, the one or more encryption-decryption algorithms can include one or more symmetric encryption algorithms (e.g., advanced encryption standard (AES), data encryption standard (DES), and/or other symmetric encryption algorithm), one or more asymmetric encryption algorithms using public and private keys (e.g., Rivest-Shamir-Adleman (RSA) and/or other asymmetric encryption algorithm), and/or other encryption-decryption algorithm.

(87) In some examples, the ITS **455** can determine certain operations (e.g., V2X-based operations) to perform based on messages received from other UEs. The operations can include safety-related and/or other operations, such as operations for road safety, traffic efficiency, infotainment, business, and/or other applications. In some examples, the operations can include causing the vehicle (e.g., the control system **452**) to perform automatic functions, such as automatic breaking, automatic steering (e.g., to maintain a heading in a particular lane), automatic lane change negotiation with other vehicles, among other automatic functions. In one illustrative example, a message can be received by the communications system **458** from another vehicle (e.g., over a PC5 interface, a DSRC interface, or other device to device direct interface) indicating that the other vehicle is coming to a sudden stop. In response to receiving the message, the ITS stack can generate a message or instruction and can send the message or instruction to the control system **452**, which can cause the control system **452** to automatically break the vehicle **404** so that it comes to a stop before making impact with the other vehicle. In other illustrative examples, the operations can include triggering display of a message alerting a driver that another vehicle is in the lane next to the vehicle, a message alerting the driver to stop the vehicle, a message alerting the driver that a pedestrian is in an upcoming cross-walk, a message alerting the driver that a toll booth is within a certain distance (e.g., within 1 mile) of the vehicle, among others.

(88) In some examples, the ITS **455** can receive a large number of messages from the other UEs (e.g., vehicles, RSUs, etc.), in which case the ITS **455** will authenticate (e.g., decode and decrypt) each of the messages and/or determine which operations to perform. Such a large number of messages can lead to a large computational load for the vehicle computing system **450**. In some cases, the large computational load can cause a temperature of the computing system **450** to increase. Rising temperatures of the components of the computing system **450** can adversely affect the ability of the computing system **450** to process the large number of incoming messages. One or more functionalities can be transitioned from the vehicle **404** to another device (e.g., a user device, a RSU, etc.) based on a temperature of the vehicle computing system **450** (or component thereof) exceeding or approaching one or more thermal levels. Transitioning the one or more functionalities can reduce the computational load on the vehicle **404**, helping to reduce the temperature of the components. A thermal load balancer can be provided that enable the vehicle computing system **450** to perform thermal based load balancing to control a processing load depending on the temperature of the computing system **450** and processing capacity of the vehicle computing system **450**.

(89) The computing system **450** further includes one or more sensor systems **456** (e.g., a first sensor system through an Nth sensor system, where N is a value equal to or greater than 0). When including multiple sensor systems, the sensor system(s) **456** can include different types of sensor systems that can be arranged on or in different parts the vehicle **404**. The sensor system(s) **456** can

include one or more camera sensor systems, LIDAR sensor systems, radio detection and ranging (RADAR) sensor systems, Electromagnetic Detection and Ranging (EmDAR) sensor systems, Sound Navigation and Ranging (SONAR) sensor systems, Sound Detection and Ranging (SODAR) sensor systems, Global Navigation Satellite System (GNSS) receiver systems (e.g., one or more Global Positioning System (GPS) receiver systems), accelerometers, gyroscopes, inertial measurement units (IMUs), infrared sensor systems, laser rangefinder systems, ultrasonic sensor systems, infrasonic sensor systems, microphones, any combination thereof, and/or other sensor systems. It should be understood that any number of sensors or sensor systems can be included as part of the computing system **450** of the vehicle **404**.

(90) While the vehicle computing system **450** is shown to include certain components and/or systems, one of ordinary skill will appreciate that the vehicle computing system **450** can include more or fewer components than those shown in FIG. **4**. For example, the vehicle computing system **450** can also include one or more input devices and one or more output devices (not shown). In some implementations, the vehicle computing system **450** can also include (e.g., as part of or separate from the control system **452**, the infotainment system **454**, the communications system **458**, and/or the sensor system(s) **456**) at least one processor and at least one memory having computer-executable instructions that are executed by the at least one processor. The at least one processor is in communication with and/or electrically connected to (referred to as being “coupled to” or “communicatively coupled”) the at least one memory. The at least one processor can include, for example, one or more microcontrollers, one or more central processing units (CPUs), one or more field programmable gate arrays (FPGAs), one or more graphics processing units (GPUs), one or more application processors (e.g., for running or executing one or more software applications), and/or other processors. The at least one memory can include, for example, read-only memory (ROM), random access memory (RAM) (e.g., static RAM (SRAM)), electrically erasable programmable read-only memory (EEPROM), flash memory, one or more buffers, one or more databases, and/or other memory. The computer-executable instructions stored in or on the at least memory can be executed to perform one or more of the functions or operations described herein.

(91) FIG. **5** illustrates an example of a computing system **570** of a user device **507**. The user device **507** is an example of a UE that can be used by an end-user. For example, the user device **507** can include a mobile phone, router, tablet computer, laptop computer, tracking device, wearable device (e.g., a smart watch, glasses, an XR device, etc.), Internet of Things (IoT) device, and/or other device used by a user to communicate over a wireless communications network. The computing system **570** includes software and hardware components that can be electrically or communicatively coupled via a bus **589** (or may otherwise be in communication, as appropriate). For example, the computing system **570** includes one or more processors **584**. The one or more processors **584** can include one or more CPUs, ASICs, FPGAs, APs, GPUs, VPUs, NSPs, microcontrollers, dedicated hardware, any combination thereof, and/or other processing device or system. The bus **589** can be used by the one or more processors **584** to communicate between cores and/or with the one or more memory devices **586**.

(92) The computing system **570** may also include one or more memory devices **586**, one or more digital signal processors (DSPs) **582**, one or more SIMs **574**, one or more modems **576**, one or more wireless transceivers **578**, an antenna **587**, one or more input devices **572** (e.g., a camera, a mouse, a keyboard, a touch sensitive screen, a touch pad, a keypad, a microphone, and/or the like), and one or more output devices **580** (e.g., a display, a speaker, a printer, and/or the like).

(93) The one or more wireless transceivers **578** can receive wireless signals (e.g., signal **588**) via antenna **587** from one or more other devices, such as other user devices, vehicles (e.g., vehicle **404** of FIG. **4** described above), network devices (e.g., base stations such as eNBs and/or gNBs, WiFI routers, etc.), cloud networks, and/or the like. In some examples, the computing system **570** can include multiple antennae. The wireless signal **588** may be transmitted via a wireless network. The wireless network may be any wireless network, such as a cellular or telecommunications network

(e.g., 3G, 4G, 5G, etc.), wireless local area network (e.g., a WiFi network), a Bluetooth™ network, and/or other network. In some examples, the one or more wireless transceivers **578** may include an RF front end including one or more components, such as an amplifier, a mixer (also referred to as a signal multiplier) for signal down conversion, a frequency synthesizer (also referred to as an oscillator) that provides signals to the mixer, a baseband filter, an analog-to-digital converter (ADC), one or more power amplifiers, among other components. The RF front-end can generally handle selection and conversion of the wireless signals **588** into a baseband or intermediate frequency and can convert the RF signals to the digital domain.

(94) In some cases, the computing system **570** can include a coding-decoding device (or CODEC) configured to encode and/or decode data transmitted and/or received using the one or more wireless transceivers **578**. In some cases, the computing system **570** can include an encryption-decryption device or component configured to encrypt and/or decrypt data (e.g., according to the AES and/or DES standard) transmitted and/or received by the one or more wireless transceivers **578**.

(95) The one or more SIMs **574** can each securely store an IMSI number and related key assigned to the user of the user device **507**. As noted above, the IMSI and key can be used to identify and authenticate the subscriber when accessing a network provided by a network service provider or operator associated with the one or more SIMs **574**. The one or more modems **576** can modulate one or more signals to encode information for transmission using the one or more wireless transceivers **578**. The one or more modems **576** can also demodulate signals received by the one or more wireless transceivers **578** in order to decode the transmitted information. In some examples, the one or more modems **576** can include a 4G (or LTE) modem, a 5G (or NR) modem, a modem configured for V2X communications, and/or other types of modems. The one or more modems **576** and the one or more wireless transceivers **578** can be used for communicating data for the one or more SIMs **574**.

(96) The computing system **570** can also include (and/or be in communication with) one or more non-transitory machine-readable storage media or storage devices (e.g., one or more memory devices **586**), which can include, without limitation, local and/or network accessible storage, a disk drive, a drive array, an optical storage device, a solid-state storage device such as a RAM and/or a ROM, which can be programmable, flash-updateable and/or the like. Such storage devices may be configured to implement any appropriate data storage, including without limitation, various file systems, database structures, and/or the like.

(97) In various aspects, functions may be stored as one or more computer-program products (e.g., instructions or code) in memory device(s) **586** and executed by the one or more processor(s) **584** and/or the one or more DSPs **582**. The computing system **570** can also include software elements (e.g., located within the one or more memory devices **586**), including, for example, an operating system, device drivers, executable libraries, and/or other code, such as one or more application programs, which may comprise computer programs implementing the functions provided by various aspects, and/or may be designed to implement methods and/or configure systems, as described herein.

(98) FIG. **6** illustrates an example 600 of wireless communication between devices based on sidelink communication, such as V2X or other D2D communication. The communication may be based on a slot structure. For example, transmitting UE **602** may transmit a transmission **614**, e.g., comprising a control channel and/or a corresponding data channel, that may be received by receiving UEs **604**, **606**, **608**. At least one UE may comprise an autonomous vehicle or an unmanned aerial vehicle. A control channel may include information for decoding a data channel and may also be used by receiving device to avoid interference by refraining from transmitting on the occupied resources during a data transmission. The number of TTIs, as well as the RBs that will be occupied by the data transmission, may be indicated in a control message from the transmitting device. The UEs **602**, **604**, **606**, **608** may each be capable of operating as a transmitting device in addition to operating as a receiving device. Thus, UEs **606**, **608** are illustrated as transmitting

transmissions **616**, **620**. The transmissions **614**, **616**, **620** (and **618** by RSU **607**) may be broadcast or multicast to nearby devices. For example, UE **614** may transmit communication intended for receipt by other UEs within a range **601** of UE **614**. Additionally/alternatively, RSU **607** may receive communication from and/or transmit communication **618** to UEs **602**, **604**, **606**, **608**. UE **602**, **604**, **606**, **608** or RSU **607** may comprise a detection component. UE **602**, **604**, **606**, **608** or RSU **607** may also comprise a BSM or mitigation component.

(99) In wireless communications, such as V2X communications, V2X entities may perform sensor sharing with other V2X entities for cooperative and automated driving. For example, with reference to diagram **700** of FIG. **7**A, the host vehicle (HV) **702** may detect a number of items within its environment. For example, the HV **702** may detect the presence of the non-V2X entity (NV) **706** at block **732**. The HV **702** may inform other entities, such as a first remote vehicle (RV1) **704** or a road side unit (RSU) **708**, about the presence of the NV **706**, if the RV1 **704** and/or the RSU **708**, by themselves, are unable to detect the NV **706**. The HV **702** informing the RV1 **704** and/or the RSU **708** about the NV **706** is a sharing of sensor information. With reference to diagram **710** of FIG. **7**B, the HV **702** may detect a physical obstacle **712**, such as a pothole, debris, or an object that may be an obstruction in the path of the HV **702** and/or RV1 **704** that has not yet been detected by RV1 **704** and/or RSU **708**. The HV **702** may inform the RV1 and/or the RSU **708** of the obstacle **712**, such that the obstacle **712** may be avoided. With reference to diagram **720** of FIG. **7**C, the HV **702** may detect the presence of a vulnerable road user (VRU) **722** and may share the detection of the VRU **722** with the RV1 **704** and the RSU **708**, in instances where the RSU **708** and/or RV1 **704** may not be able to detect the VRU **722**. With reference to diagram **730** of FIG. **7**D, the HV, upon detection of a nearby entity (e.g., NV, VRU, obstacle) may transmit a sensor data sharing message (SDSM) **734** to the RV and/or the RSU to share the detection of the entity. The SDSM **734** may be a broadcast message such that any receiving device within the vicinity of the HV may receive the message. In some instances, the shared information may be relayed to other entities, such as RVs. For example, with reference to diagram **800** of FIG. **8**, the HV **802** may detect the presence of the NV **806** and/or the VRU **822**. The HV **802** may broadcast the SDSM **810** to the RSU **808** to report the detection of NV **806** and/or VRU **822**. The RSU **808** may relay the SDSM **810** received from the HV **802** to remote vehicles such that the remote vehicles are aware of the presence of the NV **806** and/or VRU **822**. For example, the RSU **808** may transmit an SDSM **812** to the RV1 **804**, where the SDSM **812** includes information related to the detection of NV **806** and/or VRU **822**.

(100) FIG. **9** is a diagram illustrating an example of a system **900** for sensor sharing in wireless communications (e.g., C-V2X communications), in accordance with some aspects of the present disclosure. In FIG. **9**, the system **900** is shown to include a plurality of equipped (e.g., V2X capable) network devices. The plurality of equipped network devices includes vehicles (e.g., automobiles) **910***a*, **910***b*, **910***c*, **910***d*, and an RSU **905**. Also shown are a plurality of non-equipped network devices, which include a non-equipped vehicle **920**, a VRU (e.g., a bicyclist) **930**, and a pedestrian **940**. The system **900** may comprise more or less equipped network devices and/or more or less non-equipped network devices, than as shown in FIG. **9**. In addition, the system **900** may comprise more or less different types of equipped network devices (e.g., which may include equipped UEs) and/or more or less different types of non-equipped network devices (e.g., which may include non-equipped UEs) than as shown in FIG. **9**. In addition, in one or more examples, the equipped network devices may be equipped with heterogeneous capability, which may include, but is not limited to, C-V2X/DSRC capability, 4G/5G cellular connectivity, GPS capability, camera capability, radar capability, and/or LIDAR capability.

(101) The plurality of equipped network devices may be capable of performing V2X communications. In addition, at least some of the equipped network devices are configured to transmit and receive sensing signals for radar (e.g., RF sensing signals) and/or LIDAR (e.g., optical sensing signals) to detect nearby vehicles and/or objects. Additionally or alternatively, in some

cases, at least some of the equipped network devices are configured to detect nearby vehicles and/or objects using one or more cameras (e.g., by processing images captured by the one or more cameras to detect the vehicles/objects). In one or more examples, vehicles **910***a*, **910***b*, **910***c*, **910***d* and RSU **905** may be configured to transmit and receive sensing signals of some kind (e.g., radar and/or LIDAR sensing signals).

(102) In some examples, some of the equipped network devices may have higher capability sensors (e.g., GPS receivers, cameras, RF antennas, and/or optical lasers and/or optical sensors) than other equipped network devices of the system **900**. For example, vehicle **910***b* may be a luxury vehicle and, as such, have more expensive, higher capability sensors than other vehicles that are economy vehicles. In one illustrative example, vehicle **910***b* may have one or more higher capability LIDAR sensors (e.g., high capability optical lasers and optical sensors) than the other equipped network devices in the system **900**. In one illustrative example, a LIDAR of vehicle **910***b* may be able to detect a VRU (e.g., cyclist) **930** and/or a pedestrian **940** with a large degree of confidence (e.g., a seventy percent degree of confidence). In another example, vehicle **910***b* may have higher capability radar (e.g., high capability RF antennas) than the other equipped network devices in the system **900**. For instance, the radar of vehicle **910***b* may be able to detect the VRU (e.g., cyclist) **930** and/or pedestrian **940** with a degree of confidence (e.g., an eight-five percent degree of confidence). In another example, vehicle **910***b* may have higher capability camera (e.g., with higher resolution capabilities, higher frame rate capabilities, better lens, etc.) than the other equipped network devices in the system **900**.

(103) During operation of the system **900**, the equipped network devices (e.g., RSU **905** and/or at least one of the vehicles **910***a*, **910***b*, **910***c*, **910***d*) may transmit and/or receive sensing signals (e.g., RF and/or optical signals) to sense and detect vehicles (e.g., vehicles **910***a*, **910***b*, **910***c*, **910***d*, and **920**) and/or objects (e.g., VRU **930** and pedestrian **940**) located within and surrounding the road. The equipped network devices (e.g., RSU **905** and/or at least one of the vehicles **910***a*, **910***b*, **910***c*, **910***d*) may then use the sensing signals to determine characteristics (e.g., motion, dimensions, type, heading, and speed) of the detected vehicles and/or objects. The equipped network devices (e.g., RSU **905** and/or at least one of the vehicles **910***a*, **910***b*, **910***c*, **910***d*) may generate at least one vehicle-based message **915** (e.g., a C-V2X message, such as a Sensor Data Sharing Message (SDSM), a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), Collective Perception Messages (CPMs), and/or other type of message) including information related to the determined characteristics of the detected vehicles and/or objects.

(104) The vehicle-based message **915** may include information related to the detected vehicle or object (e.g., a position of the vehicle or object, an accuracy of the position, a speed of the vehicle or object, a direction in which the vehicle or object is traveling, and/or other information related to the vehicle or object), traffic conditions (e.g., low speed and/or dense traffic, high speed traffic, information related to an accident, etc.), weather conditions (e.g., rain, snow, etc.), message type (e.g., an emergency message, a non-emergency or "regular" message), etc.), road topology (line-of-sight (LOS) or non-LOS (NLOS), etc.), any combination, thereof, and/or other information. In some examples, the vehicle-based message **915** may also include information regarding the equipped network device's preference to receive vehicle-based messages from other certain equipped network devices. In some cases, the vehicle-based message **915** may include the current capabilities of the equipped network device (e.g., vehicles **910***a*, **910***b*, **910***c*, **910***d*), such as the equipped network device's sensing capabilities (which can affect the equipped network device's accuracy in sensing vehicles and/or objects), processing capabilities, the equipped network device's thermal status (which can affect the vehicle's ability to process data), and the equipped network device's state of health.

(105) In some aspects, the vehicle-based message **915** may include a dynamic neighbor list (also referred to as a Local Dynamic Map (LDM) or a dynamic surrounding map) for each of the equipped network devices (e.g., vehicles **910***a*, **910***b*, **910***c*, **910***d* and RSU **905**). For example, each

dynamic neighbor list can include a listing of all of the vehicles and/or objects that are located within a specific predetermined distance (or radius of distance) away from a corresponding equipped network device. In some cases, each dynamic neighbor list includes a mapping, which may include roads and terrain topology, of all of the vehicles and/or objects that are located within a specific predetermined distance (or radius of distance) away from a corresponding equipped network device.

(106) In some implementations, the vehicle-based message **915** may include a specific use case or safety warning, such as a do-not-pass warning (DNPW) or a forward collision warning (FCW), related to the current conditions of the equipped network device (e.g., vehicles **910***a*, **910***b*, **910***c*, **910***d*). In some examples, the vehicle-based message **915** may be in the form of a standard Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM) (e.g., SAE J3224 SDSM), and/or other format.

(107) FIG. **10** is a diagram **1000** illustrating an example of a vehicle-based message (e.g., vehicle-based message **915** of FIG. **9**), in accordance with some aspects of the present disclosure. The vehicle-based message **915** is shown as a sensor-sharing message (e.g., an SDSM), but can include a BSM, a CAM, a CPM, or other vehicle-based message as noted herein. In FIG. **10**, the vehicle-based message **915** is shown to include HostData **1020** and Detected Object Data **1010***a*, **1010***b*. The HostData **1020** of the vehicle-based message **915** may include information related to the transmitting device (e.g., the transmitting equipped network entity, such as RSU **905** or an onboard unit (OBU), such as on vehicles **910***a*, **910***b*, **910***c*, **910***d*) of the vehicle-based message **915**. The Detected Object Data **1010***a*, **1010***b* of the vehicle-based message **915** may include information related to the detected vehicle or object (e.g., static or dynamic characteristics related to the detected vehicle or object, and/or other information related to the detected vehicle or object). The Detected Object Data **1010***a*, **1010***b* may specifically include Detected Object CommonData, Detected Object VehicleData, Detected Object VRUData, Detected Obstacle ObstacleData, and Detected Object MisbehavingVehicleData.

(108) These vehicle-based messages **915** are beneficial because they can provide an awareness and understanding to the equipped network devices (e.g., vehicles **910***a*, **910***b*, **910***c*, **910***d* of FIG. **9**) of upcoming potential road dangers (e.g., unforeseen oncoming vehicles, accidents, and road conditions).

(109) FIG. **11** is a diagram illustrating an example of a system **1100** employing a road side unit (RSU) **1120** as a bridge for communications between vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f* equipped for different types of communication protocols (e.g., C-V2X and DSRC communication protocols), in accordance with some aspects of the present disclosure. In particular, in FIG. **11**, the system **1100** is shown to include a plurality of equipped (e.g., C-V2X capable and/or DSRC capable) network devices (e.g., UEs, such as vehicles). The plurality of equipped network devices includes vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f* (e.g., automobiles), and an RSU **1120**. The system **1100** may comprise more or less equipped network devices, than as shown in FIG. **11**. Also, the system **1100** may additionally comprise non-equipped network devices, such as non-equipped vehicles, VRUs (e.g., a bicyclist), and/or pedestrians. In addition, the system **1100** may comprise more or less different types of equipped network devices (e.g., which may include equipped UEs), than as shown in FIG. **11**. In addition, in one or more examples, the equipped network devices may be equipped with heterogeneous capability, which may include, but is not limited to, C-V2X and DSRC capability, 4G and 5G cellular connectivity, GPS capability, camera capability, radar capability, LIDAR capability, any other combination thereof, and/or other types of communication protocols. While examples are described herein using C-V2X and DSRC for illustrative purposes, the systems and techniques apply to other types of communications protocols, such as 4G and 5G cellular connectivity, etc.

(110) The plurality of equipped network devices may be capable of performing communications

(e.g., C-V2X and/or DSRC communications) with each other. Vehicles **1110***b*, **1110***d*, **1110***f* may be equipped to perform C-V2X communications to utilize a C-V2X sidelink PC5 interface for communications with other vehicles similarly equipped for C-V2X communications (e.g., to transmit and/or receive vehicle-based messages, such as safety and awareness messages and warnings). One of these vehicles, vehicle **1110***d*, may also be equipped for DSRC communications to utilize an 802.11p based DSRC interface for communications with other vehicles (e.g., vehicles **1110***a*, **1110***c*, **1110***e*) similarly equipped for DSRC communications (e.g., to transmit and/or receive vehicle-based messages). Vehicle **1110***d*, which may have this dual communications capability, can be said to have a heterogeneous C-V2X/DSRC communications capability. Similar to vehicle **1110***d*, the RSU **1120** may be equipped to perform both C-V2X communications to utilize a C-V2X sidelink PC5 interface for communications and DSRC communications to utilize an 802.11p based DSRC interface.

(111) Vehicles **1110***a*, **1110***c*, **1110***e* within the system **1100** may only be equipped for DSRC communications to utilize an 802.11p based DSRC interface for communications with other vehicles similarly equipped for DSRC communications (e.g., to transmit and/or receive vehicle-based messages). As such, vehicles **1110***a*, **1110***c*, **1110***e* that may only be equipped for DSRC communications may not be able to directly communicate with the vehicles **1110***b*, **1110***f* that are not equipped for DSRC communications. As such, vehicle-based messages (e.g., safety, awareness, and/or warning messages) generated by vehicles **1110***a*, **1110***c*, **1110***e* that are only be equipped for DSRC communications may not easily be received by vehicles **1110***b*, **1110***f* that may only be equipped for C-V2X communications. Similarly, vehicle-based messages (e.g., safety, awareness, and/or warning messages) generated by vehicles **1110***b*, **1110***f* that may only be equipped for C-V2X communications may not easily be received by vehicles **1110***a*, **1110***c*, **1110***e* that may only be equipped for DSRC communications.

(112) In addition, at least some of the equipped network devices (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*, and RSU **1120**) are configured to transmit and receive sensing signals for radar (e.g., RF sensing signals) and/or LIDAR (e.g., optical sensing signals) to detect nearby vehicles and/or objects. Additionally or alternatively, in some cases, at least some of the equipped network devices are configured to detect nearby vehicles and/or objects using one or more cameras (e.g., by processing images captured by the one or more cameras to detect the vehicles/objects). In one or more examples, the vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*, and RSU **1120** may be configured to transmit and receive sensing signals of some kind (e.g., radar and/or LIDAR sensing signals).

(113) During operation of the system **1100**, the equipped network devices (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*, and RSU **1120**) may transmit and/or receive sensing signals (e.g., RF and/or optical signals) to sense and detect vehicles (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*) and/or objects (e.g., RSU **1120**) located within and surrounding the road. The equipped network devices (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*, and RSU **1120**) may then use the sensing signals to determine characteristics (e.g., motion, dimensions, type, heading, and speed) of the detected vehicles and/or objects. The equipped network devices (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f* and RSU **1120**) may generate at least one vehicle-based message (e.g., vehicle-based message **915** of FIG. **10**), such as a C-V2X message, such as a Sensor Data Sharing Message (SDSM), a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), Collective Perception Messages (CPMs), and/or other type of message. The vehicle-based message(s) may include information related to the determined characteristics of the detected vehicles and/or objects.

(114) The vehicle-based message **915** may include information related to the detected vehicle or object (e.g., a position of the vehicle or object, an accuracy of the position, a speed of the vehicle or object, a direction in which the vehicle or object is traveling, and/or other information related to the vehicle or object), traffic conditions (e.g., low speed and/or dense traffic, high speed traffic,

information related to an accident, etc.), weather conditions (e.g., rain, snow, etc.), message type (e.g., an emergency message, a non-emergency or "regular" message), etc.), road topology (line-of-sight (LOS) or non-LOS (NLOS), etc.), any combination, thereof, and/or other information. In some examples, the vehicle-based message **915** may also include information regarding the equipped network device's preference to receive vehicle-based messages from other certain equipped network devices. In some cases, the vehicle-based message **915** may include the current capabilities of the equipped network device (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*), such as the equipped network device's sensing capabilities (which can affect the equipped network device's accuracy in sensing vehicles and/or objects), processing capabilities, the equipped network device's thermal status (which can affect the vehicle's ability to process data), and the equipped network device's state of health.

(115) In some aspects, the vehicle-based message **915** may include a dynamic neighbor list (also referred to as a Local Dynamic Map (LDM) or a dynamic surrounding map) for each of the equipped network devices (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*, and RSU **1120**). For example, each dynamic neighbor list can include a listing of all of the vehicles and/or objects that are located within a specific predetermined distance (or radius of distance) away from a corresponding equipped network device. In some cases, each dynamic neighbor list includes a mapping, which may include roads and terrain topology, of all of the vehicles and/or objects that are located within a specific predetermined distance (or radius of distance) away from a corresponding equipped network device.

(116) In some implementations, the vehicle-based message **915** may include a specific use case or safety warning, such as a do-not-pass warning (DNPW) or a forward collision warning (FCW), related to the current conditions of the equipped network device (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*). In some examples, the vehicle-based message **915** may be in the form of a standard Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM) (e.g., SAE J3224 SDSM), and/or other format.

(117) In one or more examples, the equipped network devices (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*, and RSU **1120**) may transmit the vehicle-based messages (e.g., vehicle-based message **915**) via message signals (e.g., message signals **1130***a*, **1130***b*, **1130***c*, **1140***a*, **1140***b*, **1140***c*, **1140***d*) to the other equipped network devices (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*, and RSU **1120**) to provide situational awareness (e.g., awareness of potential upcoming road dangers and/or obstacles) to the other equipped network devices (e.g., vehicles **1110***a*, **1110***b*, **1110***c*, **1110***d*, **1110***e*, **1110***f*, and RSU **1120**). In FIG. **11**, as well as in FIGS. **12**, **13**A, **13**B, **14**A, and **14**B, message signals (e.g., message signals **1130***a*, **1130***b*, **1130***c* of FIG. **11**) denoted by a solid line may be transmitted via C-V2X communications, and message signals (e.g., message signals **1140***a*, **1140***b*, **1140***c*, **1140***d* of FIG. **11**) denoted by a dashed line may be transmitted via C-V2X communications.

(118) For example, as shown in FIG. **11**, vehicle **1110***b* and RSU **1120** can both be transmitting message signals **1130***a* to each other via C-V2X communications, and vehicle **1110***d* and vehicle **1110***f* can both be transmitting message signals **1130***c* to each other via C-V2X communications. Also in FIG. **11**, vehicle **1110***c* and RSU **1120** can both be transmitting message signals **1140***a* to each other via DSRC communications, vehicle **1110***c* and vehicle **1110***d* can both be transmitting message signals **1140***b* to each other via DSRC communications, and vehicle **1110***c* and vehicle **1110***e* can both be transmitting message signals **1140***d* to each other via DSRC communications. In addition, vehicle **1110***d* and RSU **1120** can both be transmitting message signals **1130***b*, **1140***c* to each other via both C-V2X communications (e.g., for message signals **1130***b*) and via DSRC communications (e.g., for message signals **1140***c*).

(119) In some cases, the RSU **1120**, which may be capable of both C-V2X and DSRC communications, may operate as a bridge (or a relay) to relay message signals (e.g., which contain

vehicle-based messages) to equipped network devices that cannot directly communicate with other equipped network devices. For example, since vehicle **1110***c* is only capable of DSRC communications and vehicle **1110***b* is only capable of C-V2X communications, vehicle **1110***c* and vehicle **1110***b* cannot directly communicate vehicle-based messages with each other. In this case, the RSU **1120** can act as a relay to relay message signals to and from both vehicle **1110***c* and **1110***b*. For this example, vehicle **1110***c* may transmit message signals **1140***a* (e.g., containing vehicle-based messages from vehicle **1110***c*) to the RSU **1120** via DSRC communications. The RSU **1120** can then transmit message signals **1130***a* (e.g., the containing vehicle-based messages from vehicle **1110***c*) to vehicle **1110***b* via C-V2X communications. The use of an RSU **1120** as a bridge requires the use of many additional message signals, which can cause congestion in the channel. In addition, the RSU **1120** has the limitation that it must be within communications range of the vehicles to be able to operate as a bridge between the vehicles. Also, in FIG. **11**, every vehicle (e.g., vehicle **1110***d*) in the system that is capable of both C-V2X communications and DSRC communications may be actively transmitting via both C-V2X communications and DSRC communications. The transmission of all of these signals (e.g., transmission of excessive signals via DSRC communications) can also cause congestion in the channel.

(120) The present disclosure provides a platoon-based communication protocol interworking solution (e.g., interworking between C-V2X and DSRC protocols) to reduce the unnecessary congestion in the channel. For instance, a platoon may be formed from a grouping of vehicles capable of at least C-V2X communications, where at least one of the vehicles within the platoon is selected or determined to act as the platoon head or leader. The platoon head of the platoon can be capable of both C-V2X and DSRC communications, and can act as a bridge (e.g., similar to RSU **1120** in FIG. **11**) between the vehicles for C-V2X and DSRC communications of messages signals. When the platoon head is acting as a bridge, the other vehicles within the platoon (e.g., which may be referred to as platoon members) may cease transmitting via DSRC communications, which allows for a reduction in congestion in the channel. FIGS. **12**, **13**A, **13**B, **14**A, **14**B, **15**, and **16** discuss in detail the formation and operation of the platoon for C-V2X and DSRC communications amongst the vehicles.

(121) FIG. **12** is a diagram of the disclosed system **1200** for a platoon-based C-V2X and DSRC interworking solution, where some vehicles **1110***b*, **1110***d*, **1110***f* within the system **1200** are collaboratively negotiating (e.g., via signals **1230***a*, **1230***b*) the formation of a platoon, in accordance with some aspects of the present disclosure. Specifically, in FIG. **12**, the system **1200** may include a plurality of equipped (e.g., C-V2X capable and/or DSRC capable) network devices (e.g., UEs, which may be in the form of vehicles). The plurality of equipped network devices may include vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f* (e.g., automobiles). The system **1200** may include more or less equipped network devices, than as shown in FIG. **12**. The system **1200** may also include non-equipped network devices, such as non-equipped vehicles, VRUs (e.g., a bicyclist), and/or pedestrians. Also, the system **1200** may include more or less different types of equipped network devices (e.g., which may include equipped UEs), than as shown in FIG. **12**. In addition, in one or more aspects, the equipped network devices may be equipped with heterogeneous capability, which may include, but is not limited to, C-V2X/DSRC capability, 4G/5G cellular connectivity, GPS capability, camera capability, radar capability, and/or LIDAR capability.

(122) The plurality of equipped network devices may be capable of performing communications (e.g., C-V2X and/or DSRC communications) with each other. Vehicles **1210***b*, **1210***d*, **1210***f* may be equipped to perform C-V2X communications to utilize a C-V2X sidelink PC5 interface for communications with other vehicles similarly equipped for C-V2X communications (e.g., to transmit and/or receive vehicle-based messages, such as safety, awareness, and/or warning messages). Vehicle **1210***d*, may also be equipped for DSRC communications to utilize an 802.11p based DSRC interface for communications with other vehicles (e.g., vehicles **1210***a*, **1210***c*, **1210***e*)

similarly equipped for DSRC communications (e.g., to transmit and/or receive vehicle-based messages). Vehicle **1210***d* with this dual communications capability is said to have a heterogeneous C-V2X/DSRC communications capability.

(123) Vehicles **1210***a*, **1210***c*, **1210***e* may only be equipped for DSRC communications to utilize an 802.11p based DSRC interface for communications with other vehicles similarly equipped for DSRC communications (e.g., to transmit and/or receive vehicle-based messages). As such, vehicles **1210***a*, **1210***c*, **1210***e* that may only be equipped for DSRC communications may not be able to directly communicate with the vehicles **1210***b*, **1210***f* that are not equipped for DSRC communications. As such, vehicle-based messages (e.g., safety, awareness, and/or warning messages) generated by vehicles **1210***a*, **1210***c*, **1210***e* that may only be equipped for DSRC communications may not easily be received by vehicles **1210***b*, **1210***f*, which may only be equipped for C-V2X communications. Vehicle-based messages (e.g., safety, awareness, and/or warning messages) generated by vehicles **1210***b*, **1210***f* that may only be equipped for C-V2X communications may not easily be received by vehicles **1210***a*, **1210***c*, **1210***e* that may only be equipped for DSRC communications.

(124) At least some of the equipped network devices (e.g., vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) may be configured to transmit and receive sensing signals for radar (e.g., RF sensing signals) and/or LIDAR (e.g., optical sensing signals) to detect nearby vehicles (e.g., vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) and/or objects (e.g., VRUs and pedestrians). In some cases, at least some of the equipped network devices (e.g., vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) may be configured to detect nearby vehicles and/or objects using one or more cameras (e.g., by processing images captured by the one or more cameras to detect the vehicles/objects). In one or more examples, the vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f* may be configured to transmit and receive sensing signals of some kind (e.g., radar and/or LIDAR sensing signals).

(125) During operation of the system **1200**, the equipped network devices (e.g., vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) may transmit and/or receive sensing signals (e.g., RF and/or optical signals) to sense and detect vehicles (e.g., vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) and/or objects (e.g., VRUs and pedestrians) located within and surrounding the road. The equipped network devices (e.g., vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) may then use the sensing signals to determine characteristics (e.g., motion, dimensions, type, heading, and speed) of the detected vehicles (e.g. vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) and/or objects (e.g., VRUs and pedestrians). The equipped network devices (e.g. vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) may generate at least one vehicle-based message (e.g., vehicle-based message **915** of FIG. **10**), such as a C-V2X message, such as a Sensor Data Sharing Message (SDSM), a Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), Collective Perception Messages (CPMs), and/or other type of message. The vehicle-based message(s) may include information related to the determined characteristics of the detected vehicles and/or objects.

(126) In one or more examples, the vehicle-based message (e.g., vehicle-based message **915** of FIG. **10**) can include information related to the detected vehicle or object (e.g., a position of the vehicle or object, an accuracy of the position, a speed of the vehicle or object, a direction in which the vehicle or object is traveling, and/or other information related to the vehicle or object), traffic conditions (e.g., low speed and/or dense traffic, high speed traffic, information related to an accident, etc.), weather conditions (e.g., rain, snow, etc.), message type (e.g., an emergency message, a non-emergency or “regular” message), etc.), road topology (line-of-sight (LOS) or non-LOS (NLOS), etc.), any combination, thereof, and/or other information. In one or more examples, the vehicle-based message (e.g., vehicle-based message **915** of FIG. **10**) can also include information regarding the equipped network device's preference to receive vehicle-based messages from other certain equipped network devices. In some cases, the vehicle-based message (e.g., vehicle-based message **915** of FIG. **10**) can include the current capabilities of the equipped network

device (e.g. vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*), such as the equipped network device's sensing capabilities (which can affect the equipped network device's accuracy in sensing vehicles and/or objects), processing capabilities, the equipped network device's thermal status (which can affect the vehicle's ability to process data), and the equipped network device's state of health.
(127) In some examples, the vehicle-based message (e.g., vehicle-based message **915** of FIG. **10**) can include a dynamic neighbor list (e.g., a Local Dynamic Map (LDM) or a dynamic surrounding map) for each of the equipped network devices (e.g. vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*). For example, each dynamic neighbor list can include a listing of all of the vehicles (e.g. vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*) and/or objects (e.g., VRUs and pedestrians) that are located within a specific predetermined distance (or radius of distance) away from a corresponding equipped network device. In some cases, each dynamic neighbor list includes a mapping, which may include roads and terrain topology, of all of the vehicles and/or objects that are located within a specific predetermined distance (or radius of distance) away from a corresponding equipped network device.
(128) In some aspects, the vehicle-based message (e.g., vehicle-based message **915** of FIG. **10**) can include a specific use case or safety warning, such as a do-not-pass warning (DNPW) or a forward collision warning (FCW), related to the current conditions of the equipped network device (e.g. vehicles **1210***a*, **1210***b*, **1210***c*, **1210***d*, **1210***e*, **1210***f*). In some examples, the vehicle-based message (e.g., vehicle-based message **915** of FIG. **10**) may be in the form of a standard Basic Safety Message (BSM), a Cooperative Awareness Message (CAM), a Collective Perception Message (CPM), a Sensor Data Sharing Message (SDSM) (e.g., SAE J3224 SDSM), and/or other format.
(129) In one or more examples, during operation of the disclosed system **1200**, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*), which are located within a grouping and are equipped for at least C-V2X communications, may transmit platoon formation signals **1230***a*, **1230***b* (e.g., via C-V2X communications) to each other regarding the formation of a platoon. In some examples, after the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) have received any C-V2X signals from at least one vehicle, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) may then transmit the platoon formation signals **1230***a*, **1230***b* (e.g., via C-V2X communications) to that vehicle(s).
(130) The platoon may include vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) that are within a communication range from each other and/or may include vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) that are located proximate to one another within a zone of a designated size (e.g., a zone having a predetermined radius). At least one vehicle (e.g., vehicles **1210***d*) within the platoon needs to be capable of both C-V2X and DSRC communications to act as a bridge for C-V2X and DSRC communications.
(131) After the platoon has been formed, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) can collectively determine (e.g., via the platoon formation signals **1230***a*, **1230***b*) which vehicle in the platoon can be designated as the platoon head. There can be multiple factors that may be weighed by the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) regarding the determination of the platoon head (e.g., vehicle **1210***d*), which may include, but are not limited to, the communications capability of the vehicles (e.g., the platoon head needs to be capable of both C-V2X and DSRC communications in order to operate as a bridge for C-V2X and DSRC communications amongst vehicles), the location of the vehicles (e.g., it can be advantageous to have the platoon head located centrally within the platoon so that the platoon head can always be within communications range of the other vehicles within the platoon), and the processing capability of the vehicles (e.g., the platoon head will need to be able to process all of the received vehicle-based messages to generate a summary message). The remaining vehicles within the platoon that are not designated as the platoon head can be referred to as platoon members.
(132) In the example shown in FIG. **12**, after the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) form a platoon with themselves, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) may select vehicle

1210*d* to be the platoon head because vehicle **1210***d* can be capable of both C-V2X and DSRC communications, and is centrally located within the platoon vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*). After the platoon is formed and a platoon head has been chosen, any platoon members (vehicles) within the platoon that are capable of C-V2X and DSRC communications my cease transmitting via DSRC communications, which will allow for a reduction in congestion in the channel. In one or more examples, the platoon members (vehicles) within the platoon that are capable of C-V2X and DSRC communications may utilize an adaptive algorithm to control their transmission via DSRC communications to reduce congestion in the channel, when needed.

(133) In one or more examples, after the platoon has been formed, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) within the platoon (e.g., including the platoon head and the platoon members) may continue to monitor the vehicles within the platoon as well as continue to monitor vehicles not within the platoon that are located proximate to the platoon. For example, the vehicles within the platoon may monitor the location of the vehicles within the platoon regarding their change in proximity to the other vehicles within the platoon. If a vehicle within the platoon moves away from (e.g., is no longer located proximate to) the other vehicles in the platoon, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) in the platoon may decide that the vehicle should no longer be included in the platoon. For another example, the vehicles in the platoon may monitor the vehicles not in the platoon, which are located proximate to the platoon, regarding their location with respect to the platoon and their communications capability (e.g., C-V2X communications capability). If the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) in the platoon determine that the vehicle is located close enough to the platoon to be within communications range of vehicles in the platoon, and the vehicle is capable of at least C-V2X communications, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) in the platoon may decide to include the vehicle within the platoon.

(134) In one or more examples, after the platoon has been formed, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) within the platoon may continue to monitor the vehicle (e.g., vehicle **1210***d*) chosen has the platoon head regarding the vehicle's (e.g., vehicle **1210***d*) communications capability (e.g., the platoon head needs to be capable of both C-V2X and DSRC communications in order to operate as a bridge for C-V2X and DSRC communications amongst vehicles), location (e.g., it can be advantageous to have the platoon head located centrally within the platoon so that the platoon head can always be within communications range of the other vehicles within the platoon), and/or processing capability (e.g., the platoon head will need to be able to process received vehicle-based messages to generate a summary message). If the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) in the platoon determine that that the vehicle (e.g., **1210***d*) designated as the platoon head is no longer centrally located within the platoon such that the vehicle can no longer receive communications from all of the vehicles in the platoon and/or is no longer capable of processing all of the received vehicle-based messages, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) in the platoon may choose another vehicle in the platoon to operate as the platoon head instead of the vehicle (e.g., vehicle **1110***d*) previously designated as the platoon head.

(135) In one or more examples, after the platoon has been formed, the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) within the platoon may continue to monitor the size of the platoon. For example, if the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) within the platoon decide that the platoon has become too large (e.g., the number of vehicles in the platoon is larger than a predetermined maximum number of vehicles for the platoon and/or the number of vehicles in the platoon requires too great of a processing capability for the platoon head to process all of the vehicle-based messages from all of the vehicles in the platoon), the vehicles (e.g., vehicles **1210***b*, **1210***d*, **1210***f*) in the platoon may decide to split the platoon into multiple separate platoons (e.g., two separate platoons), which will each have their own designated platoon head.

(136) FIGS. **13**A and **13**B are diagrams of the disclosed system **1200** for a platoon-based C-V2X and DSRC interworking solution that together illustrate an example of a platoon head (e.g., vehicle **1210***d*) operating as a bridge for communications between vehicles (e.g., vehicles **1210***a*, **1210***c*,

1210*e*, **1210***f*) equipped for different types of communications (e.g., C-V2X and DSRC communications), in accordance with some aspects of the present disclosure. In particular, in FIG. **13**A, during operation of the disclosed system **1200**, after the platoon has been formed and a platoon head (e.g., vehicle **1210***d*) has been chosen, vehicles **1210***a*, **1210***c*, **1210***e*, which are equipped only for DSRC communications and are not members of the platoon, may transmit message signals **1340***a*, **1340***b*, **1340***c* (e.g., which may contain vehicle-based messages, such as safety, awareness, and/or warning messages) via DSRC communications to the vehicle **1210***d*, which may be operating as the platoon head. After vehicle **1210***d* receives all of the message signals **1340***a*, **1340***b*, **1340***c* from the vehicles **1210***a*, **1210***c*, **1210***e*, the vehicle **1210***d* (e.g., operating as the platoon head) may process the vehicle-based messages contained within the message signals **1340***a*, **1340***b*, **1340***c* to summarize the information contained within the vehicle-based messages. After the vehicle **1210***d* has summarized the information contained within the vehicle-based messages, the vehicle **1210***d* (e.g., operating as the platoon head) may generate a summary message containing the summary of the information. In one or more examples, the platoon head (e.g., vehicle **1210***d*) may add the summary message at the end of a vehicle-based message (e.g., at the end of a BSM).
(137) In FIG. **13**B, during operation of the disclosed system **1200**, after the vehicle **1210***d* (e.g., operating as the platoon head) has generated the summary message, the vehicle **1210***d* may transmit message signals **1330***a*, **1330***b* (e.g., via C-V2X communications), which contain the summary message, to the platoon members (e.g., vehicles **1210***b*, **1210***f*) to provide situational awareness to the platoon members (e.g., vehicles **1210***b*, **1210***f*).
(138) FIGS. **14**A and **14**B are diagrams of the disclosed system **1200** for a platoon-based C-V2X and DSRC interworking solution that together illustrate another example of a platoon head (e.g., **1210***d*) operating as a bridge for communications between vehicles (e.g., vehicles **1210***a*, **1210***c*, **1210***e*, **1210***f*) equipped for different types of communications (e.g., C-V2X and DSRC communications), in accordance with some aspects of the present disclosure. In particular, in FIG. **14**A, during operation of the disclosed system **1200**, after the platoon has been formed and a platoon head (e.g., vehicle **1210***d*) has been chosen, vehicles **1210***b*, **1210***f*, which are members of the platoon, may transmit message signals **1430***a*, **1430***b* (e.g., which may contain vehicle-based messages, such as safety, awareness, and/or warning messages) via C-V2X communications to the vehicle **1210***d*, which may be operating as the platoon head. After vehicle **1210***d* receives all of the message signals **1430***a*, **1430***b* from the vehicles **1210***b*, **1210***f*, the vehicle **1210***d*, operating as the platoon head, can process the vehicle-based messages contained within the message signals **1430***a*, **1430***b* to summarize the information contained within the vehicle-based messages. After the vehicle **1210***d* has summarized the information contained within the vehicle-based messages, the vehicle **1210***d*, operating as the platoon head, can generate a summary message containing the summary of the information. In one or more examples, the platoon head (e.g., vehicle **1210***d*) may add the summary message at the end of a vehicle-based message (e.g., at the end of a BSM).
(139) In FIG. **14**B, during operation of the disclosed system **1200**, after the vehicle **1210***d* (e.g., operating as the platoon head) has generated the summary message, the vehicle **1210***d* may transmit message signals **1440***a*, **1440***b*, **1440***c* (e.g., via DSRC communications), which contain the summary message, to the vehicles **1210***a*, **1210***c*, **1210***e* to provide situational awareness to the vehicles **1210***a*, **1210***c*, **1210***e*.
(140) FIG. **15** is a flow diagram illustrating an example of a process **1500** for forming a platoon, in accordance with some aspects of the present disclosure. At block **1510**, a vehicle (e.g., **1210***d* of FIGS. **12**, **13**A, **13**B, **14**A, and **14**B), which may be equipped for C-V2X communications, may be started (e.g., the vehicle may be turned on, such as the vehicle's engine can be started). At block **1520**, the vehicle (e.g., **1210***d*), which may be equipped for both C-V2X communications and DSRC communications, can enable its radio such that the vehicle can be capable of transmitting and receiving both C-V2X and DSRC communications (e.g., turn on C-V2X and DSRC

communications).
(141) At block **1530**, the vehicle (e.g., vehicle **1210***d*), which may be equipped for C-V2X communications, may determine whether the vehicle has received any vehicle-based messages via C-V2X communications from any vehicles (e.g., vehicles **1210***b*, **1210***f*) that are located within communications range of the vehicle (e.g., **1210***d*). If the vehicle (e.g., vehicle **1210***d*) determines that it did not receive any vehicle-based messages via C-V2X communications, at block **1540**, the vehicle can simply continue to enable its radio such that the vehicle can be capable of transmitting and receiving both C-V2X and DSRC communications (e.g., turn on C-V2X and DSRC communications). However, if the vehicle (e.g., vehicle **1210***d*) determines that it did receive at least one vehicle-based message via C-V2X communications, at block **1550**, the vehicle may communicate with the vehicle(s) (e.g., vehicles **1210***b*, **1210***f*) that transmitted the vehicle-based message(s) to the vehicle (e.g., vehicle **1210***d*) regarding forming a platoon. However, if a platoon has already been formed (e.g., a platoon formed including vehicles **1210***b*, **1210***f*), the vehicle (e.g., vehicle **1210***d*) may join the already-formed platoon.
(142) Then, at block **1560**, the vehicle (e.g., **1210***d*) may determine whether a platoon has been formed, for example, where the platoon may include the vehicle (e.g., **1210***d*) and the vehicle(s) (e.g., vehicles **1210***b*, **1210***f*) that transmitted the vehicle-based message(s) to the vehicle (e.g., vehicle **1210***d*). If the vehicle (e.g., **1210***d*) determines that a platoon has not been formed, at block **1570**, the vehicle can simply continue to enable its radio such that the vehicle can be capable of transmitting and receiving both C-V2X and DSRC communications (e.g., turn on C-V2X and DSRC communications). However, if the vehicle (e.g., **1210***d*) determines that a platoon has been formed, at block **1580**, DSRC platoon mode can be enabled.
(143) During DSRC platoon mode, the vehicle (e.g., **1210***d*) designated as the platoon head of the platoon may enable its radio such that the vehicle (e.g., **1210***d*) can be capable of transmitting and receiving both C-V2X and DSRC communications (e.g., turn on C-V2X and DSRC communications). Also during DSRC platoon mode, the other vehicles in the platoon (e.g., the vehicles in the platoon that are not the platoon head), which may be referred to as platoon members, that are capable of transmitting and receiving both C-V2X and DSRC communications may partially disable their radio such that they may only receive via DSRC communications (e.g., may not be able to transmit via DSRC communications), but may continue to receive both C-V2X communications and DSRC communications.
(144) FIG. **16** is a flow chart illustrating an example of a process **1600** for wireless communications. The process **1600** can be performed by a network device (e.g., a UE such as vehicle, a mobile device including a mobile phone or other mobile device, a network-connected watch, an extended reality device (e.g., a virtual reality (VR) device, an augmented reality (AR) device, a mixed reality (MR) device, etc.) or by a component or system (e.g., a chipset) of the network device. The operations of the process **1600** may be implemented as software components that are executed and run on one or more processors (e.g., processor **1710** of FIG. **17** or other processor(s)). Further, the transmission and reception of signals by the wireless communications device in the process **1600** may be enabled, for example, by one or more antennas and/or one or more transceivers (e.g., wireless transceiver(s)).
(145) At block **1610**, the network device (or component thereof) may receive a plurality of messages using a first type of communication protocol. The network device is part of a platoon including a plurality of network devices (e.g., a plurality of UEs, such as the vehicles shown in FIG. **12**). In some cases, the plurality of messages are vehicle-based messages, such as one or more Sensor Data Sharing Messages (SDSMs), one or more Basic Safety Messages (BSMs), one or more Cooperative Awareness Messages (CAMs), one or more Collective Perception Messages (CPMs), and/or other types of vehicle-based messages.
(146) In some aspects, the platoon is formed based on at least a location of the network device and a respective location of each network device of the plurality of network devices. In some cases, the

network device is a platoon head of the platoon based on at least the location of the network device relative to the respective location of each network device of the plurality of network devices and relative to at least one other network device not included in the platoon. In some cases, the network device is the platoon head of the platoon further based on at least one capability of the network device. For instance, the at least one capability may include a capability of the network device to communicate using a first type of communication protocol (e.g., a dedicated short-range communications (DSRC) protocol) and the second type of communication protocol (e.g., a cellular vehicle-to-everything (C-V2X) protocol).

(147) At block **1620**, the network device (or component thereof) may generate a summary message summarizing the plurality of messages. For instance, the summary message may include message content (e.g., a collision warning, an indicating a vehicle is breaking, an indication of a location of a vehicle) from each of the messages. In some cases, common message information from multiple messages can be consolidated into a single item of information (e.g., two messages indicating a location of a pedestrian may be consolidated into a single item of message content indicating the location of the pedestrian). In some cases, the summary message is transmitted at an end of a vehicle-based message (e.g., in a final or last field of the vehicle-based message), which may include a BSM, SDSM, CAM, or other type of vehicle-based message.

(148) At block **1630**, the network device (or component thereof) may transmit the summary message using a second type of communication protocol. In some cases, the first type of communication protocol is a dedicated short-range communications (DSRC) protocol and the second type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol. In other cases, the first type of communication protocol is C-V2X, and the second type of communication protocol is the DSRC protocol. The first and second communication protocols can include any other examples of communications protocols, the first communication protocol being different the second communication protocol.

(149) In some aspects, the network device (or component thereof) may receive the plurality of messages from at least one network device not included in the platoon. In such aspects, the network device (or component thereof) may transmit the summary message to the plurality of network devices included in the platoon. In some cases, the at least one network device not included in the platoon is configured to communicate using the first type of communication protocol (e.g., the DSRC protocol) and is not configured to communicate using the second type of communication protocol (e.g., the C-V2X protocol), the plurality of network devices included in the platoon are configured to communicate using at least the second type of communication protocol (e.g., the C-V2X protocol), and the network device is configured to communicate using the first type of communication protocol (e.g., the DSRC protocol) and the second type of communication protocol (e.g., the C-V2X protocol).

(150) In some aspects, the network device (or component thereof) may receive the plurality of messages from the plurality of network devices included in the platoon. In such aspects, the network device (or component thereof) may transmit the summary message to at least one network device not included in the platoon. In some cases, the at least one network device not included in the platoon is configured to communicate using the second type of communication protocol (e.g., the C-V2X protocol) and is not configured to communicate using the first type of communication protocol (e.g., the DSRC protocol), the plurality of network devices included in the platoon are configured to communicate using the first type of communication protocol (e.g., the DSRC protocol) and in some instances the second type of communication protocol (e.g., the C-V2X protocol), and the network device is configured to communicate using the first type of communication protocol (e.g., the DSRC protocol) and the second type of communication protocol (e.g., the C-V2X protocol).

(151) In some aspects, the network device (or component thereof) may determine a first location of the network device. The network device (or component thereof) may determine a respective

location and at least one respective capability of each network device of the plurality of network devices included in the platoon. The network device (or component thereof) may further determine a second location of at least one network device not included in the platoon. The network device (or component thereof) may then determine a second network device from the plurality of network devices as a platoon head of the platoon based on the first location of the network device, the respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon, and the second location of the at least one network device not included in the platoon. In some cases, the network device (or component thereof) may transmit, to the second network device, a message including information indicative of the second network device as the platoon head (e.g., instructing the second network device that it has become the platoon head).
(152) FIG. **17** is a block diagram illustrating an example of a computing system **1700**, which may be employed by the disclosed system for a platoon-based C-V2X and DSRC interworking solution, in accordance with some aspects of the present disclosure. In particular, FIG. **17** illustrates an example of computing system **1700**, which can be for example any computing device making up internal computing system, a remote computing system, a camera, or any component thereof in which the components of the system are in communication with each other using connection **1705**. Connection **1705** can be a physical connection using a bus, or a direct connection into processor **1710**, such as in a chipset architecture. Connection **1705** can also be a virtual connection, networked connection, or logical connection.
(153) In some aspects, computing system **1700** is a distributed system in which the functions described in this disclosure can be distributed within a datacenter, multiple data centers, a peer network, etc. In some aspects, one or more of the described system components represents many such components each performing some or all of the function for which the component is described. In some aspects, the components can be physical or virtual devices.
(154) Example system **1700** includes at least one processing unit (CPU or processor) **1710** and connection **1705** that communicatively couples various system components including system memory **1715**, such as read-only memory (ROM) **1720** and random access memory (RAM) **1725** to processor **1710**. Computing system **1700** can include a cache **1712** of high-speed memory connected directly with, in close proximity to, or integrated as part of processor **1710**.
(155) Processor **1710** can include any general purpose processor and a hardware service or software service, such as services **1732**, **1734**, and **1736** stored in storage device **1730**, configured to control processor **1710** as well as a special-purpose processor where software instructions are incorporated into the actual processor design. Processor **1710** may essentially be a completely self-contained computing system, containing multiple cores or processors, a bus, memory controller, cache, etc. A multi-core processor may be symmetric or asymmetric.
(156) To enable user interaction, computing system **1700** includes an input device **1745**, which can represent any number of input mechanisms, such as a microphone for speech, a touch-sensitive screen for gesture or graphical input, keyboard, mouse, motion input, speech, etc. Computing system **1700** can also include output device **1735**, which can be one or more of a number of output mechanisms. In some instances, multimodal systems can enable a user to provide multiple types of input/output to communicate with computing system **1700**.
(157) Computing system **1700** can include communications interface **1740**, which can generally govern and manage the user input and system output. The communication interface may perform or facilitate receipt and/or transmission wired or wireless communications using wired and/or wireless transceivers, including those making use of an audio jack/plug, a microphone jack/plug, a universal serial bus (USB) port/plug, an Apple™ Lightning™ port/plug, an Ethernet port/plug, a fiber optic port/plug, a proprietary wired port/plug, 3G, 4G, 5G and/or other cellular data network wireless signal transfer, a Bluetooth™ wireless signal transfer, a Bluetooth™ low energy (BLE) wireless signal transfer, an IBEACON™ wireless signal transfer, a radio-frequency identification (RFID)

wireless signal transfer, near-field communications (NFC) wireless signal transfer, dedicated short range communication (DSRC) wireless signal transfer, 802.11 Wi-Fi wireless signal transfer, wireless local area network (WLAN) signal transfer, Visible Light Communication (VLC), Worldwide Interoperability for Microwave Access (WiMAX), Infrared (IR) communication wireless signal transfer, Public Switched Telephone Network (PSTN) signal transfer, Integrated Services Digital Network (ISDN) signal transfer, ad-hoc network signal transfer, radio wave signal transfer, microwave signal transfer, infrared signal transfer, visible light signal transfer, ultraviolet light signal transfer, wireless signal transfer along the electromagnetic spectrum, or some combination thereof.

(158) The communications interface **1740** may also include one or more range sensors (e.g., LIDAR sensors, laser range finders, RF radars, ultrasonic sensors, and infrared (TR) sensors) configured to collect data and provide measurements to processor **1710**, whereby processor **1710** can be configured to perform determinations and calculations needed to obtain various measurements for the one or more range sensors. In some examples, the measurements can include time of flight, wavelengths, azimuth angle, elevation angle, range, linear velocity and/or angular velocity, or any combination thereof. The communications interface **1740** may also include one or more Global Navigation Satellite System (GNSS) receivers or transceivers that are used to determine a location of the computing system **1700** based on receipt of one or more signals from one or more satellites associated with one or more GNSS systems. GNSS systems include, but are not limited to, the US-based GPS, the Russia-based Global Navigation Satellite System (GLONASS), the China-based BeiDou Navigation Satellite System (BDS), and the Europe-based Galileo GNSS. There is no restriction on operating on any particular hardware arrangement, and therefore the basic features here may easily be substituted for improved hardware or firmware arrangements as they are developed.

(159) Storage device **1730** can be a non-volatile and/or non-transitory and/or computer-readable memory device and can be a hard disk or other types of computer readable media which can store data that are accessible by a computer, such as magnetic cassettes, flash memory cards, solid state memory devices, digital versatile disks, cartridges, a floppy disk, a flexible disk, a hard disk, magnetic tape, a magnetic strip/stripe, any other magnetic storage medium, flash memory, memristor memory, any other solid-state memory, a compact disc read only memory (CD-ROM) optical disc, a rewritable compact disc (CD) optical disc, digital video disk (DVD) optical disc, a blu-ray disc (BDD) optical disc, a holographic optical disk, another optical medium, a secure digital (SD) card, a micro secure digital (microSD) card, a Memory Stick® card, a smartcard chip, a EMV chip, a subscriber identity module (SIM) card, a mini/micro/nano/pico SIM card, another integrated circuit (IC) chip/card, random access memory (RAM), static RAM (SRAM), dynamic RAM (DRAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash EPROM (FLASHEPROM), cache memory (e.g., Level 1 (L1) cache, Level 2 (L2) cache, Level 3 (L3) cache, Level 4 (L4) cache, Level 5 (L5) cache, or other (L #) cache), resistive random-access memory (RRAM/ReRAM), phase change memory (PCM), spin transfer torque RAM (STT-RAM), another memory chip or cartridge, and/or a combination thereof.

(160) The storage device **1730** can include software services, servers, services, etc., that when the code that defines such software is executed by the processor **1710**, it causes the system to perform a function. In some aspects, a hardware service that performs a particular function can include the software component stored in a computer-readable medium in connection with the necessary hardware components, such as processor **1710**, connection **1705**, output device **1735**, etc., to carry out the function. The term “computer-readable medium” includes, but is not limited to, portable or non-portable storage devices, optical storage devices, and various other mediums capable of storing, containing, or carrying instruction(s) and/or data. A computer-readable medium may include a non-transitory medium in which data can be stored and that does not include carrier

waves and/or transitory electronic signals propagating wirelessly or over wired connections. Examples of a non-transitory medium may include, but are not limited to, a magnetic disk or tape, optical storage media such as compact disk (CD) or digital versatile disk (DVD), flash memory, memory or memory devices. A computer-readable medium may have stored thereon code and/or machine-executable instructions that may represent a procedure, a function, a subprogram, a program, a routine, a subroutine, a module, a software package, a class, or any combination of instructions, data structures, or program statements. A code segment may be coupled to another code segment or a hardware circuit by passing and/or receiving information, data, arguments, parameters, or memory contents. Information, arguments, parameters, data, etc. may be passed, forwarded, or transmitted via any suitable means including memory sharing, message passing, token passing, network transmission, or the like.

(161) Specific details are provided in the description above to provide a thorough understanding of the aspects and examples provided herein, but those skilled in the art will recognize that the application is not limited thereto. Thus, while illustrative aspects of the application have been described in detail herein, it is to be understood that the inventive concepts may be otherwise variously embodied and employed, and that the appended claims are intended to be construed to include such variations, except as limited by the prior art. Various features and aspects of the above-described application may be used individually or jointly. Further, aspects can be utilized in any number of environments and applications beyond those described herein without departing from the broader scope of the specification. The specification and drawings are, accordingly, to be regarded as illustrative rather than restrictive. For the purposes of illustration, methods were described in a particular order. It should be appreciated that in alternate aspects, the methods may be performed in a different order than that described.

(162) For clarity of explanation, in some instances the present technology may be presented as including individual functional blocks comprising devices, device components, steps or routines in a method embodied in software, or combinations of hardware and software. Additional components may be used other than those shown in the figures and/or described herein. For example, circuits, systems, networks, processes, and other components may be shown as components in block diagram form in order not to obscure the aspects in unnecessary detail. In other instances, well-known circuits, processes, algorithms, structures, and techniques may be shown without unnecessary detail in order to avoid obscuring the aspects.

(163) Further, those of skill in the art will appreciate that the various illustrative logical blocks, modules, circuits, and algorithm steps described in connection with the aspects disclosed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, but such implementation decisions should not be interpreted as causing a departure from the scope of the present disclosure.

(164) Individual aspects may be described above as a process or method which is depicted as a flowchart, a flow diagram, a data flow diagram, a structure diagram, or a block diagram. Although a flowchart may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed, but could have additional steps not included in a figure. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc. When a process corresponds to a function, its termination can correspond to a return of the function to the calling function or the main function.

(165) Processes and methods according to the above-described examples can be implemented using

computer-executable instructions that are stored or otherwise available from computer-readable media. Such instructions can include, for example, instructions and data which cause or otherwise configure a general purpose computer, special purpose computer, or a processing device to perform a certain function or group of functions. Portions of computer resources used can be accessible over a network. The computer executable instructions may be, for example, binaries, intermediate format instructions such as assembly language, firmware, source code. Examples of computer-readable media that may be used to store instructions, information used, and/or information created during methods according to described examples include magnetic or optical disks, flash memory, USB devices provided with non-volatile memory, networked storage devices, and so on.
(166) In some aspects the computer-readable storage devices, mediums, and memories can include a cable or wireless signal containing a bitstream and the like. However, when mentioned, non-transitory computer-readable storage media expressly exclude media such as energy, carrier signals, electromagnetic waves, and signals per se.
(167) Those of skill in the art will appreciate that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof, in some cases depending in part on the particular application, in part on the desired design, in part on the corresponding technology, etc.
(168) The various illustrative logical blocks, modules, and circuits described in connection with the aspects disclosed herein may be implemented or performed using hardware, software, firmware, middleware, microcode, hardware description languages, or any combination thereof, and can take any of a variety of form factors. When implemented in software, firmware, middleware, or microcode, the program code or code segments to perform the necessary tasks (e.g., a computer-program product) may be stored in a computer-readable or machine-readable medium. A processor(s) may perform the necessary tasks. Examples of form factors include laptops, smart phones, mobile phones, tablet devices or other small form factor personal computers, personal digital assistants, rackmount devices, standalone devices, and so on. Functionality described herein also can be embodied in peripherals or add-in cards. Such functionality can also be implemented on a circuit board among different chips or different processes executing in a single device, by way of further example.
(169) The instructions, media for conveying such instructions, computing resources for executing them, and other structures for supporting such computing resources are example means for providing the functions described in the disclosure.
(170) The techniques described herein may also be implemented in electronic hardware, computer software, firmware, or any combination thereof. Such techniques may be implemented in any of a variety of devices such as general purposes computers, wireless communication device handsets, or integrated circuit devices having multiple uses including application in wireless communication device handsets and other devices. Any features described as modules or components may be implemented together in an integrated logic device or separately as discrete but interoperable logic devices. If implemented in software, the techniques may be realized at least in part by a computer-readable data storage medium comprising program code including instructions that, when executed, performs one or more of the methods, algorithms, and/or operations described above. The computer-readable data storage medium may form part of a computer program product, which may include packaging materials. The computer-readable medium may comprise memory or data storage media, such as random access memory (RAM) such as synchronous dynamic random access memory (SDRAM), read-only memory (ROM), non-volatile random access memory (NVRAM), electrically erasable programmable read-only memory (EEPROM), FLASH memory, magnetic or optical data storage media, and the like. The techniques additionally, or alternatively,

may be realized at least in part by a computer-readable communication medium that carries or communicates program code in the form of instructions or data structures and that can be accessed, read, and/or executed by a computer, such as propagated signals or waves.
(171) The program code may be executed by a processor, which may include one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, an application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Such a processor may be configured to perform any of the techniques described in this disclosure. A general-purpose processor may be a microprocessor; but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. Accordingly, the term “processor,” as used herein may refer to any of the foregoing structure, any combination of the foregoing structure, or any other structure or apparatus suitable for implementation of the techniques described herein.
(172) One of ordinary skill will appreciate that the less than (“<”) and greater than (“>”) symbols or terminology used herein can be replaced with less than or equal to (“<”) and greater than or equal to (“>”) symbols, respectively, without departing from the scope of this description.
(173) Where components are described as being “configured to” perform certain operations, such configuration can be accomplished, for example, by designing electronic circuits or other hardware to perform the operation, by programming programmable electronic circuits (e.g., microprocessors, or other suitable electronic circuits) to perform the operation, or any combination thereof.
(174) The phrase “coupled to” or “communicatively coupled to” refers to any component that is physically connected to another component either directly or indirectly, and/or any component that is in communication with another component (e.g., connected to the other component over a wired or wireless connection, and/or other suitable communication interface) either directly or indirectly.
(175) Claim language or other language reciting “at least one of” a set and/or “one or more” of a set indicates that one member of the set or multiple members of the set (in any combination) satisfy the claim. For example, claim language reciting “at least one of A and B” or “at least one of A or B” means A, B, or A and B. In another example, claim language reciting “at least one of A, B, and C” or “at least one of A, B, or C” means A, B, C, or A and B, or A and C, or B and C, or A and B and C. The language “at least one of” a set and/or “one or more” of a set does not limit the set to the items listed in the set. For example, claim language reciting “at least one of A and B” or “at least one of A or B” can mean A, B, or A and B, and can additionally include items not listed in the set of A and B.
(176) Illustrative aspects of the disclosure include: Aspect 1. A method for wireless communications at network device, the method comprising: receiving, at the network device, a plurality of messages using a first type of communication protocol, wherein the UE is part of a platoon including a plurality of UEs; generating, at the UE, a summary message summarizing the plurality of messages; and transmitting the summary message using a second type of communication protocol. Aspect 2. The method of Aspect 1, wherein the network device is a vehicle, and wherein the plurality of network devices includes a plurality of vehicles. Aspect 3. The method of any of Aspects 1 to 2, further comprising: receiving the plurality of messages from at least one network device not included in the platoon. Aspect 4. The method of any of Aspects 1 to 3, further comprising: transmitting the summary message to the plurality of network devices included in the platoon. Aspect 5. The method of any of Aspects 1 to 4, wherein: the at least one network device not included in the platoon is configured to communicate using the first type of communication protocol and is not configured to communicate using the second type of communication protocol; the plurality of network devices included in the platoon are configured to communicate using at least the second type of communication protocol; and the network device is

configured to communicate using the first type of communication protocol and the second type of communication protocol. Aspect 6. The method of any of Aspects 1 to 5, wherein the first type of communication protocol is a dedicated short-range communications (DSRC) protocol, and the second type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol. Aspect 7. The method of any of Aspects 1 to 6, further comprising: receiving the plurality of messages from the plurality of network devices included in the platoon. Aspect 8. The method of any of Aspects 1 to 7, further comprising: transmitting the summary message to at least one network device not included in the platoon. Aspect 9. The method of any of Aspects 1 to 8, wherein: the at least one network device not included in the platoon is configured to communicate using the second type of communication protocol and is not configured to communicate using the first type of communication protocol; the plurality of network devices included in the platoon are configured to communicate using at least the first type of communication protocol; and the network device is configured to communicate using the first type of communication protocol and the second type of communication protocol. Aspect 10. The method of any of Aspects 1 to 9, wherein the first type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol, and the second type of communication protocol is a dedicated short-range communications (DSRC) protocol. Aspect 11. The method of any of Aspects 1 to 10, wherein the platoon is formed based on at least a location of the network device and a respective location of each network device of the plurality of network devices. Aspect 12. The method of any of Aspects 1 to 11, wherein the network device is a platoon head of the platoon based on at least the location of the network device relative to the respective location of each network device of the plurality of network devices and relative to at least one other network device not included in the platoon. Aspect 13. The method of any of Aspects 1 to 12, wherein the network device is the platoon head of the platoon further based on at least one capability of the network device. Aspect 14. The method of any of Aspects 1 to 13, wherein the at least one capability includes a capability to communicate using the first type of communication protocol and the second type of communication protocol. Aspect 15. The method of any of Aspects 1 to 14, further comprising: determining a first location of the network device; determining a respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon; determining a second location of at least one network device not included in the platoon; and determining a second network device from the plurality of network devices as a platoon head of the platoon based on the first location of the network device, the respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon, and the second location of the at least one network device not included in the platoon. Aspect 16. The method of any of Aspects 1 to 15, further comprising: transmitting, to the second network device, a message including information indicative of the second network device as the platoon head. Aspect 17. The method of any of Aspects 1 to 16, wherein the plurality of messages are vehicle-based messages. Aspect 18. The method of any of Aspects 1 to 17, wherein the vehicle-based messages comprise at least one of a Sensor Data Sharing Messages (SDSM), a Basic Safety Messages (BSM), a Cooperative Awareness Messages (CAM), or a Collective Perception Messages (CPM). Aspect 19. The method of any of Aspects 1 to 18, wherein the summary message is transmitted at an end of a vehicle-based message. Aspect 20. An apparatus for wireless communications, comprising: at least one memory; and at least one processor coupled to at least one memory and configured to: receive a plurality of messages using a first type of communication protocol, wherein the apparatus is part of a platoon including a plurality of network devices; generate a summary message summarizing the plurality of messages; and output the summary message for transmission using a second type of communication protocol. Aspect 21. The apparatus of Aspect 20, wherein the apparatus is a computing device of a vehicle, and wherein the plurality of network devices includes a plurality of vehicles. Aspect 22. The apparatus of any of Aspects 20 to 21, wherein the at least one processor is configured to: receive the plurality of messages from at least one network device not included in the platoon. Aspect 23. The apparatus of any of Aspects 20 to 22, wherein the at least one processor is configured to: output the summary message for transmission to the plurality of network devices included in the platoon. Aspect 24. The apparatus of any of Aspects 20 to 23, wherein: the at least one network device not included in the platoon is configured to communicate using the first type of communication protocol and is not configured to communicate using the second type of communication protocol; the plurality of network devices included in the platoon are configured to communicate using at least the second type of communication protocol; and the apparatus is configured to communicate using the first type of communication protocol and the second type of communication protocol. Aspect 25. The apparatus of any of Aspects 20 to 24, the first type of communication protocol is a dedicated short-range communications (DSRC) protocol, and the second type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol. Aspect 26. The apparatus of any of Aspects 20 to 25, wherein the at least one processor is configured to: receive the plurality of messages from the plurality of network devices included in the platoon. Aspect 27. The apparatus of any of Aspects 20 to 26, wherein the at least one processor is configured to: output the summary message for transmission to at least one network device not included in the platoon. Aspect 28. The apparatus of any of Aspects 20 to 27, wherein: the at least one network device not included in the platoon is configured to communicate using the second type of communication protocol and is not configured to communicate using the first type of communication protocol; the plurality of network devices included in the platoon are configured to communicate using at least the first type of communication protocol; and the apparatus is configured to communicate using the first type of communication protocol and the second type of communication protocol. Aspect 29. The apparatus of any of Aspects 20 to 28, wherein the first type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol, and the second type of communication protocol is a dedicated short-range communications (DSRC) protocol. Aspect 30. The apparatus of any of Aspects 20 to 29, wherein the platoon is formed based on at least a location of the apparatus and a respective location of each network device of the plurality of network devices. Aspect 31. The apparatus of any of Aspects 20 to 30, wherein the apparatus is a platoon head of the platoon based on at least the location of the apparatus relative to the respective location of each network device of the plurality of network devices and relative to at least one other network device not included in the platoon. Aspect 32. The apparatus of any of Aspects 20 to 31, wherein the apparatus is the platoon head of the platoon further based on at least one capability of the apparatus. Aspect 33. The apparatus of any of Aspects 20 to 32, wherein the at least one capability includes a capability to communicate using the first type of communication protocol and the second type of communication protocol. Aspect 34. The apparatus of any of Aspects 20 to 33, wherein the at least one processor is configured to: determine a first location of the apparatus; determine a respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon; determine a second location of at least one network device not included in the platoon; and determine a network device from the plurality of network devices as a platoon head of the platoon based on the first location of the apparatus, the respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon, and the second location of the at least one network device not included in the platoon. Aspect 35. The apparatus of any of Aspects 20 to 34, wherein the at least one processor is configured to: output, for transmission to the network device, a message including information indicative of the network device as the platoon head. Aspect 36. The apparatus of any of Aspects 20 to 35, wherein the plurality of messages are vehicle-based messages. Aspect 37. The apparatus of any of Aspects 20 to 36, wherein the vehicle-based messages comprise at least one of a Sensor Data Sharing Messages (SDSM), a Basic Safety Messages (BSM), a Cooperative Awareness Messages (CAM), or a Collective Perception Messages (CPM). Aspect 38. The apparatus of any of Aspects 20 to 37, wherein the summary message is transmitted at an end of a vehicle-based message. Aspect 39. A non-transitory computer-readable

medium is provided that has stored thereon instructions that, when executed by one or more processors, cause the one or more processors to perform operations according to any of Aspects 1 to 38. Aspect 40. An apparatus for wireless communications, the apparatus comprising one or more means for performing operations according to any of Aspects 1 to 38.
(177) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean "one and only one" unless specifically so stated, but rather "one or more."

Claims

1. A method for wireless communications at a network device, the method comprising: determining the network device is selected as a platoon head of a platoon including a plurality of network devices based at least on the network device having a capability of communicating using a first type of communication protocol and a second type of communication protocol; receiving, at the network device from at least one network device not included in the platoon, a plurality of messages using the first type of communication protocol, wherein the at least one network device not included in the platoon is configured to communicate using the first type of communication protocol and is not configured to communicate using the second type of communication protocol; generating, at the network device, a summary message summarizing information included in the plurality of messages received from the at least one network device not included in the platoon using the first type of communication protocol; and transmitting, to the plurality of network devices in the platoon, the summary message using the second type of communication protocol.
2. The method of claim 1, wherein the network device is a vehicle, and wherein the plurality of network devices includes a plurality of vehicles.
3. The method of claim 1, wherein: the plurality of network devices included in the platoon are configured to communicate using at least the second type of communication protocol.
4. The method of claim 3, wherein the first type of communication protocol is a dedicated short-range communications (DSRC) protocol, and the second type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol.
5. The method of claim 1, further comprising: receiving an additional plurality of messages from the plurality of network devices included in the platoon; and generating, at the network device, an additional summary message summarizing information included in the additional plurality of messages received from the plurality of network devices included in the platoon.
6. The method of claim 5, further comprising: transmitting the additional summary message to at least one additional network device not included in the platoon.
7. The method of claim 6, wherein: the at least one additional network device not included in the platoon is configured to communicate using the second type of communication protocol and is not configured to communicate using the first type of communication protocol; and the plurality of network devices included in the platoon are configured to communicate using at least the first type of communication protocol.
8. The method of claim 7, wherein the first type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol, and the second type of communication protocol is a dedicated short-range communications (DSRC) protocol.
9. The method of claim 1, wherein the platoon is formed based on at least a location of the network device and a respective location of each network device of the plurality of network devices.
10. The method of claim 9, wherein the network device is selected as the platoon head of the platoon further based on at least the location of the network device relative to the respective

location of each network device of the plurality of network devices and relative to at least one other network device not included in the platoon.
11. The method of claim 1, further comprising: determining a first location of the network device; determining a respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon; determining a second location of at least one network device not included in the platoon; and determining a second network device from the plurality of network devices as a platoon head of the platoon based on the first location of the network device, the respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon, and the second location of the at least one network device not included in the platoon.
12. The method of claim 11, further comprising: transmitting, to the second network device, a message including information indicative of the second network device as the platoon head.
13. The method of claim 1, wherein the plurality of messages are vehicle-based messages.
14. The method of claim 13, wherein the vehicle-based messages comprise at least one of a Sensor Data Sharing Messages (SDSM), a Basic Safety Messages (BSM), a Cooperative Awareness Messages (CAM), or a Collective Perception Messages (CPM).
15. An apparatus for wireless communications, comprising: at least one memory; and at least one processor coupled to at least one memory and configured to: determine the apparatus is selected as a platoon head of a platoon including a plurality of network devices based at least on the apparatus having a capability of communicating using a first type of communication protocol and a second type of communication protocol; receive, from at least one network device not included in the platoon, a plurality of messages using the first type of communication protocol, wherein the at least one network device not included in the platoon is configured to communicate using the first type of communication protocol and is not configured to communicate using the second type of communication protocol; generate a summary message summarizing information included in the plurality of messages received from the at least one network device not included in the platoon using the first type of communication protocol; and output, to the plurality of network devices in the platoon, the summary message for transmission using the second type of communication protocol.
16. The apparatus of claim 15, wherein the apparatus is a computing device of a vehicle, and wherein the plurality of network devices includes a plurality of vehicles.
17. The apparatus of claim 15, wherein: the plurality of network devices included in the platoon are configured to communicate using at least the second type of communication protocol.
18. The apparatus of claim 17, wherein the first type of communication protocol is a dedicated short-range communications (DSRC) protocol, and the second type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol.
19. The apparatus of claim 15, wherein the at least one processor is configured to: receive an additional plurality of messages from the plurality of network devices included in the platoon; and generate an additional summary message summarizing information included in the additional plurality of messages received from the plurality of network devices included in the platoon.
20. The apparatus of claim 19, wherein the at least one processor is configured to: output the additional summary message for transmission to at least one additional network device not included in the platoon.
21. The apparatus of claim 20, wherein: the at least one additional network device not included in the platoon is configured to communicate using the second type of communication protocol and is not configured to communicate using the first type of communication protocol; and the plurality of network devices included in the platoon are configured to communicate using at least the first type of communication protocol.
22. The apparatus of claim 21, wherein the first type of communication protocol is a cellular vehicle-to-everything (C-V2X) protocol, and the second type of communication protocol is a

dedicated short-range communications (DSRC) protocol.
23. The apparatus of claim 15, wherein the platoon is formed based on at least a location of the apparatus and a respective location of each network device of the plurality of network devices.
24. The apparatus of claim 23, wherein the apparatus is selected as the platoon head of the platoon further based on at least the location of the apparatus relative to the respective location of each network device of the plurality of network devices and relative to at least one other network device not included in the platoon.
25. The apparatus of claim 15, wherein the at least one processor is configured to: determine a first location of the apparatus; determine a respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon; determine a second location of at least one network device not included in the platoon; and determine a network device from the plurality of network devices as a platoon head of the platoon based on the first location of the apparatus, the respective location and at least one respective capability of each network device of the plurality of network devices included in the platoon, and the second location of the at least one network device not included in the platoon.
26. The apparatus of claim 25, further comprising: transmitting, to the network device, a message including information indicative of the network device as the platoon head.
27. The apparatus of claim 15, wherein the plurality of messages are vehicle-based messages.